Unit 51 Metals, alloys and other synthetic materials in modern life

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

51.1 Introduction

We have studied about metals in Topic 3. In this unit, we are going to study about metals, alloys and the differences in their properties.

In metals the particles are arranged in a very regular way, giving rise to crystals. We are going to explore the crystal structures of metals.



Fig. 51.1 These oranges are packed in the same way as atoms in metallic crystals

51.2 Metallic crystal structures: close-packed structures

We can treat the metal atoms as spheres of the same size. In the following activity, you are going to build models of the close-packed structures of metallic crystals.

Activity 51.1

Building models of the close-packed structures of metallic crystals.

Reference Website
Refer to the following websites for animations on close-packed structures:
- http://departments.kings.edu/chemlab/animation/clospack.html (accessed May 2015)
- http://www.doitpoms.ac.uk/tlplib/crystallography3/packing.php (accessed May 2015)

From the above activity, you can see that there are two possible close-packed structures for metallic crystals: **hexagonal close-packed** (h.c.p.) **structure** and **cubic close-packed** (c.c.p.) **structure**.

Fig. 51.2 shows a close-packed layer of atoms. Each atom is in contact with six others. The atom marked 'a' has six other atoms in the same layer in contact with it. Fig. 51.2 also emphasizes the hexagonal arrangement of atoms within the layer.

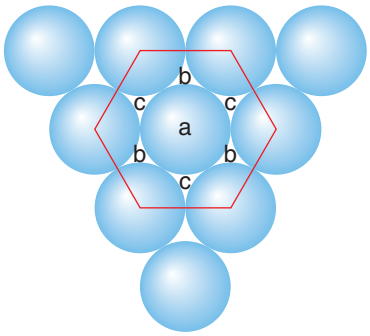


Fig. 51.2 A layer of close-packed atoms

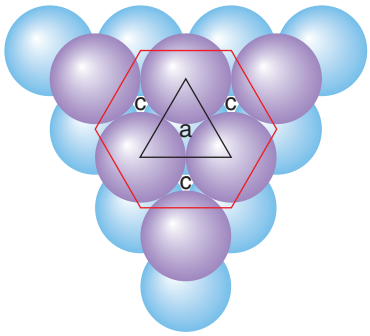


Fig. 51.3 First and second layers of close-packed atoms

Around each first-layer atom there are six depressions. These depressions are marked alternately 'b' and 'c' around the atom 'a'. Fig. 51.3 shows a second layer of atoms in place. The atoms pack as closely as possible to those in the first layer by 'sitting' in the depressions between atoms in the first layer. If the depressions marked 'b' are used, those marked 'c' cannot be.

Fig. 51.3 also shows the two types of depression available for atoms in the third layer. One type of depression, labelled 'a', lies directly above the centre of atom 'a' in the first layer. The other type of depression, labelled 'c', lies directly above a hole in the first layer.

The hexagonal close-packed structure

We can add a third layer of atoms to Fig. 51.3 in two distinct ways. In Fig. 51.4, the orange sphere is part of the third layer whose atoms are directly above the atoms of the first layer. In this case, the orange sphere occupies an 'a' type depression. We denote this type of arrangement in which the alternate layers have a similar positioning of atoms as 'ababab' because the alternate layers of atoms occupy either the 'a' depressions or the 'b' depressions. This arrangement is known as hexagonal close-packing.

hexagonal close-packed structure 六方緊密裝填結構
cubic close-packed structure 立方緊密裝填結構

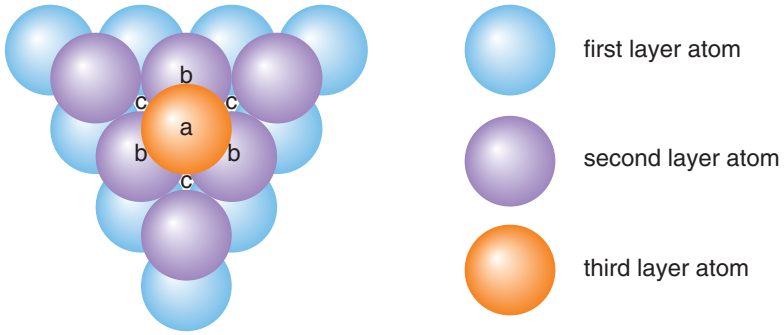


Fig. 51.4 Three layers of atoms in a hexagonal close-packing arrangement (showing abab pattern)

The diagrams in Fig. 51.5 show the arrangement of atoms in a hexagonal close-packed structure. Magnesium and zinc have hexagonal close-packed structures.

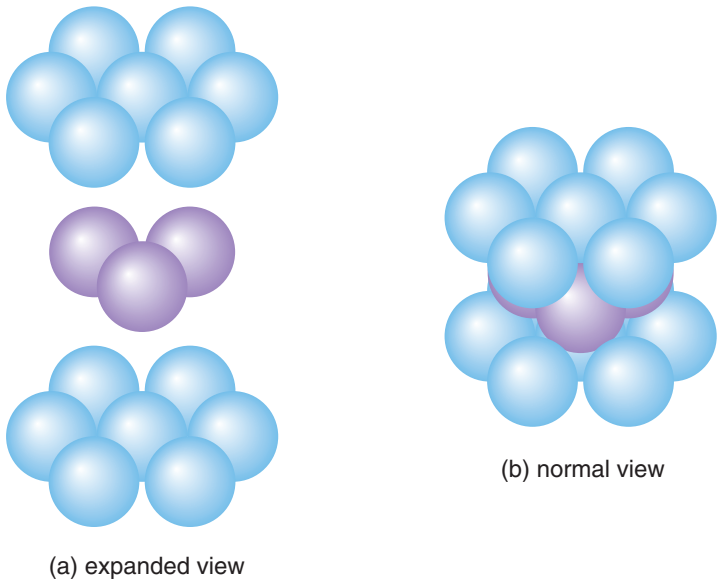


Fig. 51.5 Arrangement of atoms in a hexagonal close-packed structure

The cubic close-packed structure

Fig. 51.6 shows another close-packed arrangement in which atoms in the third layer occupy the 'c' depressions, above neither the first layer nor the second layer atoms.

Reference Website

Refer to the following websites for animations on a cubic close-packed structure:

- http://users.ece.gatech.edu/~alan/index_files/Visualization.htm (accessed May 2015)
- http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/sphere10.swf (accessed May 2015)

This type of arrangement in which the first layer of atoms occupy the 'a' depressions, the second layer of atoms occupy the 'b' depressions and the third layer of atoms occupy the 'c' depressions can be repeated to give a sequence of layers denoted as 'abcabc'. This arrangement is known as cubic close-packing.

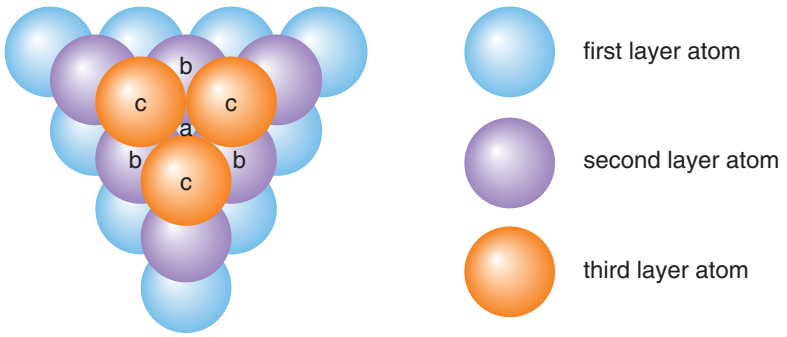
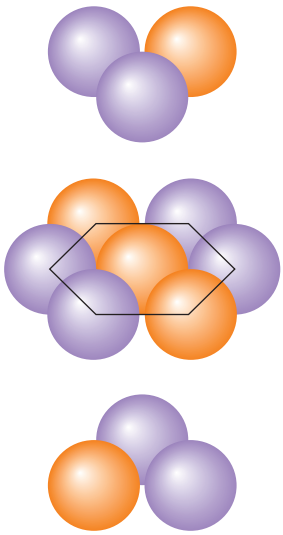
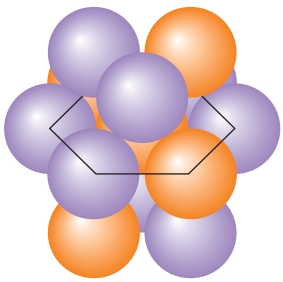


Fig. 51.6 Three layers of atoms in a cubic close-packing arrangement (showing abcabc pattern)

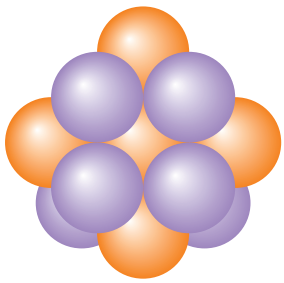
In Fig. 51.6, spheres of different colours are used to represent atoms of the different layers and they are viewed from the top. The same close packing structure is shown in Fig. 51.7(a) below, but this time spheres of different colours are used for atoms within a layer in order to emphasize the cubic close-packing structure. Fig. 51.7(c) which shows the cubic structure clearly is obtained by rotating Fig. 51.7(b) through 60°.



(a) expanded view



(b) normal view



(c) rotation of the normal view to show the cubic structure more clearly

Fig. 51.7 Arrangement of atoms in a cubic close-packed structure

Aluminium, copper, lead, silver, gold and platinum have cubic close-packed structures.

51.3 Open structure — the body-centred cubic structure

Reference Website

Refer to the following websites for animations on a body-centred cubic structure:

- http://departments.kings.edu/chemlab/animation/bcc.html (accessed May 2015)
- http://users.ece.gatech.edu/~alan/index_files/Visualization.htm (accessed May 2015)
- http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/sphere8.swf (accessed May 2015)

Some metals do not have their atoms arranged in a close-packed structure. These metals have a **body-centred cubic** (b.c.c) **structure**.

Fig. 51.8 shows a layer of atoms in a body-centred cubic structure. Atoms do not touch each other and form a square-packed layer.

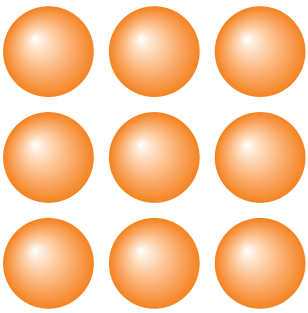


Fig. 51.8 A layer of atoms in a body-centred cubic structure

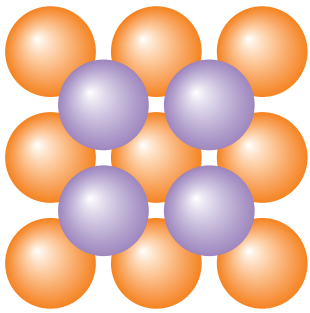


Fig. 51.9 First and second layers of atoms in a body-centred cubic structure

Atoms in the second layer sit in the depressions between the atoms in the first layer, as shown in the Fig. 51.9.

Atoms in the third layer sit in the depressions in the second layer and atoms in the fourth layer sit in the depressions in the third layer, and so on. The result is a structure in which the odd-numbered layers of atoms are identical and the even-numbered layers are identical. This repeating structure is known as body-centred cubic structure.

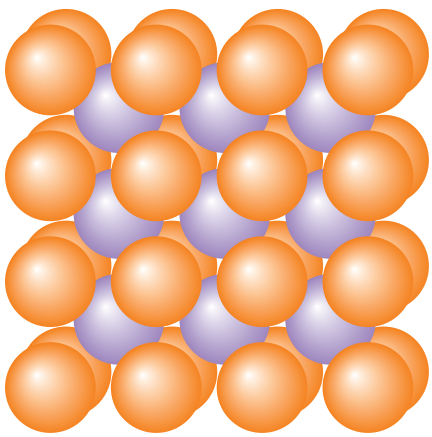


Fig. 51.10 The third layer of atoms is in registry with the first layer in a body-centred cubic structure

body-centred cubic structure 體心立方結構

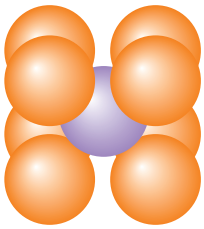


Fig. 51.11 In a body-centred cubic structure, each atom touches four atoms in the layer above and four atoms in the layer below

This structure is called body-centred cubic because each atom touches four atoms in the layer above and four more in the layer below, arranged towards the corners of a cube (Fig. 51.11). The b.c.c. arrangement does not allow the atoms to pack together as closely as the h.c.p. or c.c.p. arrangement.

All the alkali metals, iron and manganese have body-centred cubic structures.

The close-packed systems account for about 50 metals, and the body-centred system for about 20 metals. There is no obvious relation between the structural type of a metal and its position in the periodic table.

51.4 Unit cells

The body-centred cubic structure

Fig. 51.12b shows the reduced sphere view of a body-centred cubic structure. Each atom in this structure has a coordination number of 8 because it is immediately surrounded by eight others.

AL 1999 Paper 2 Q2(c)(i)(I)

The **coordination number** is defined as the number of atoms (or ions) immediately surrounding an atom (or ion) in a crystal lattice.

The **unit cell** is the simplest arrangement of atoms (or ions) which when repeated will reproduce the whole structure. N1

Reference Website
Refer to the following website for an animation on unit cell:
http://www.doitpoms.ac.uk/tlplib/crystallography3/unit_cell.php
(accessed May 2015)

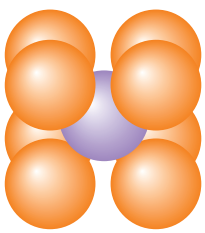
N1 Students should be able to give a precise definition of the term 'unit cell'.

N2 Students should be able to produce good drawings of different unit cells.

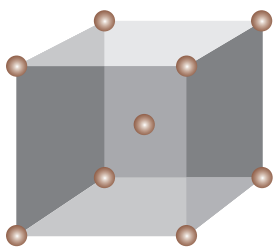
AL 2009 Paper 1 Q2(a)(i)

AL 2003 Paper 1 Q4(b)

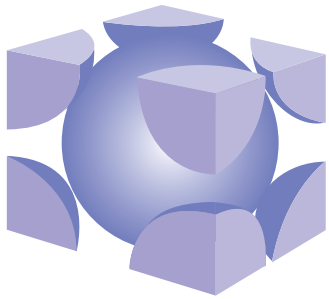
DSE 2015 Paper 2 Q2(a)(i)(1)



(a) normal view



(b) reduced sphere view



(c) unit cell N2

DSE 2012 Paper 2 Q2(c)(i)(1)

DSE 2014 Paper 2 Q2(a)(i)(2)

Fig. 51.12 Reduced sphere view and the unit cell of a body-centred cubic structure

coordination number 配位數 unit cell 晶胞

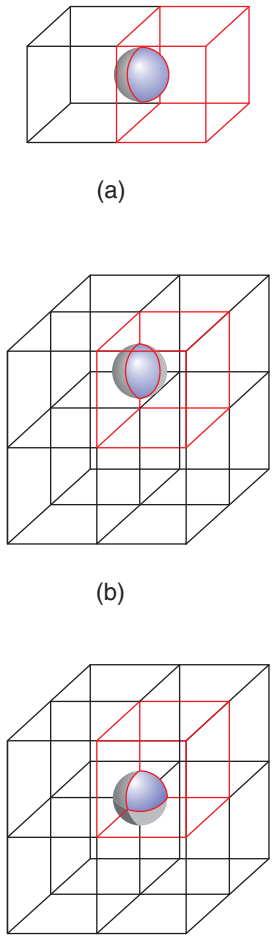


Fig. 51.13 (a) An atom at a face is shared by 2 unit cells
(b) An atom at an edge is shared by 4 unit cells
(c) An atom at a corner is shared by 8 unit cells

N3 Examination questions often ask students to calculate the number of atoms present in one unit cell of a metal.

Fig. 51.12c shows the unit cell of a body-centred cubic structure. Notice that the unit cell is a cube. We can build up the whole crystal structure by packing the little cubic unit cells together.

Consider the extent to which the various atoms in a unit cell are shared between adjacent cells:

- an atom fully inside a cell is not shared, giving 1 atom per cell;
- an atom at a face is shared by 2 cells, giving $\frac{1}{2}$ atom per cell;
- an atom at an edge is shared by 4 cells, giving $\frac{1}{4}$ atom per cell;
- an atom at a corner is shared by 8 cells, giving $\frac{1}{8}$ atom per cell.

Examine the unit cell of the b.c.c. structure. There is one atom in the centre of the body of the unit cell plus eight atoms at the corners of the cube.

$\therefore$ number of atoms per cell = $1 + \left(8 \times \frac{1}{8}\right) = 2$ N3

DSE 2012 Paper 2 Q2(c)(i)(2)

The cubic close-packed structure

In a cubic close-packed structure, any one atom touches 12 others — six in its own layer, three in the layer above and three in the layer below (Fig. 51.14). Hence the coordination number of an atom in a cubic close-packed structure is 12.

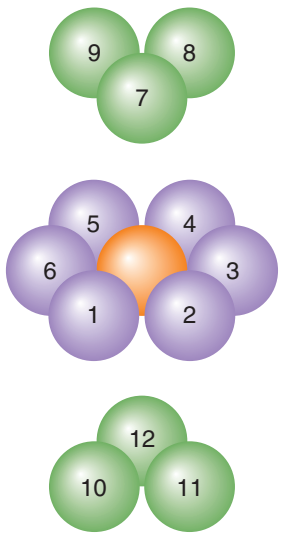
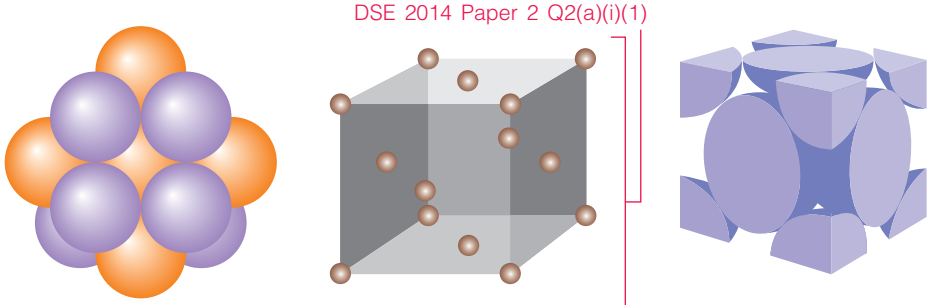


Fig. 51.14 Coordination number of an atom in a cubic close-packed structure is 12

Figs. 51.15b – c show the reduced sphere view of a cubic close-packed structure and the unit cell of the structure. This is a **face-centred cubic unit cell**. There are atoms at all the corners and the centres of all the faces of the cube.

DSE 2014 Paper 2 Q2(a)(i)(1)



(a) normal view (b) reduced sphere view (c) unit cell

Fig. 51.15 Reduced sphere view and the unit cell of a cubic close-packed structure

We can calculate the number of atoms per unit cell as follows:

- one atom at the centre of each of the six faces, giving $6 \times \frac{1}{2}$ atoms;
- one atom at each of the eight corners, giving $8 \times \frac{1}{8}$ atoms.

$$\therefore \text{ number of atoms per cell } = \left[\left(6 \times \frac{1}{2} \right) + \left(8 \times \frac{1}{8} \right) \right]$$

$$= 4$$ N4

DSE 2015 Paper 2 Q2(a)(i)(2)

N4 Refer to page T45.

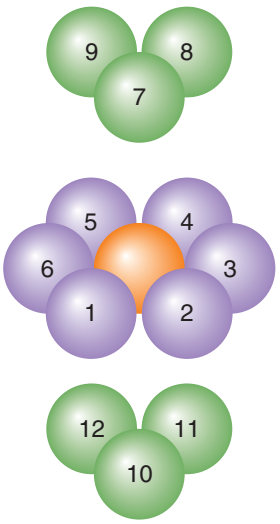


Fig. 51.16 Coordination number of an atom in a hexagonal close-packed structure is 12

The hexagonal close-packed structure

In a hexagonal close-packed structure, any one atom touches 12 others — six in its own layer, three in the layer above and three in the layer below (Fig. 51.16). Hence the coordination number of an atom in a hexagonal close-packed structure is also 12.

It is easy to understand why metals pack in hexagonal or cubic close-packed structures. Besides using space as efficiently as possible, these structures also have the largest possible coordination numbers, which allows each metal atom to form bonds to the largest number of neighbouring metal atoms.

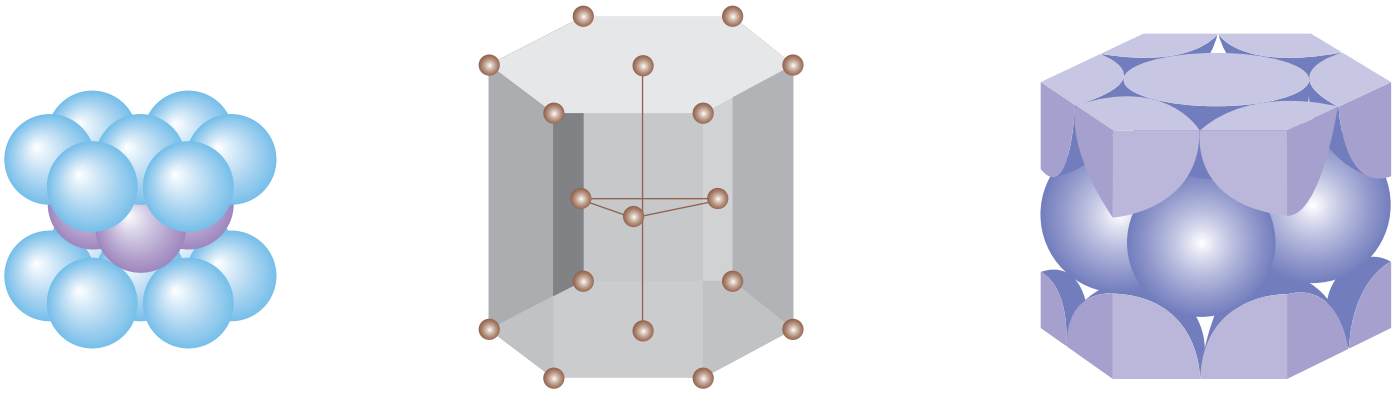
Table 51.1 summarizes the coordination number for the atoms in the three structures.

face-centred cubic unit cell 面心立方晶胞

Table 51.1 Coordination numbers for the atoms in the three different structures

Structure	Stacking pattern	Coordination number of atom
Body-centred cubic	odd-numbered layers of atoms are identical; even-numbered layers of atoms are identical	8
Cubic close-packed	abcabcabc	12
Hexagonal close-packed	abababab	12

Figs. 51.17b – c show the reduced sphere view of a hexagonal close-packed structure and the unit cell of the structure.



(a) normal view (b) reduced sphere view (c) unit cell

Fig. 51.17 Reduced sphere view and the unit cell of a hexagonal close-packed structure

We can calculate the number of atoms per unit cell as follows:

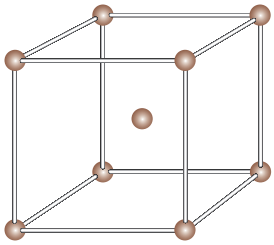
- each of the three atoms in the middle layer contributes 1 atom, giving 3 atoms;
- one atom at the centre of each of the top and bottom layers, giving $2 \times \frac{1}{2}$ atom;
- one atom at each of the 12 corners, giving $12 \times \frac{1}{6}$ atoms.

$$\therefore \quad \text{number of atoms per cell} = \left[3 + \left(2 \times \frac{1}{2} \right) + \left(12 \times \frac{1}{6} \right) \right] = 6$$

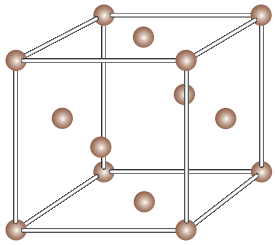
Example

51.1 Examining the crystalline forms of iron

There are several crystalline forms of iron. Below 906 °C, iron exists in the α-form. Above this temperature, iron changes to the γ-form. The unit cells of the two forms are shown below:



α-form



γ-form

a) Name the structures of the two forms.

b) Describe, with explanation, the change in volume that would occur when a piece of iron changes from the α-form to the γ-form at 906 °C.

AL 2004 Paper 2 Q4(d)(ii)

Solution

a) α-form has a body-centred cubic structure.

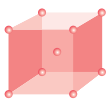
γ-form has a cubic close-packed structure.

AL 2004 Paper 2 Q4(d)(ii)

b) Atoms are more closely packed in a cubic close-packed structure than a body-centred cubic structure. Thus, when the iron changes from the α-form to the γ-form, there is a contraction in volume.

Checkpoint

1 Sodium has a body-centred cubic structure while magnesium has a hexagonal close-packed structure.

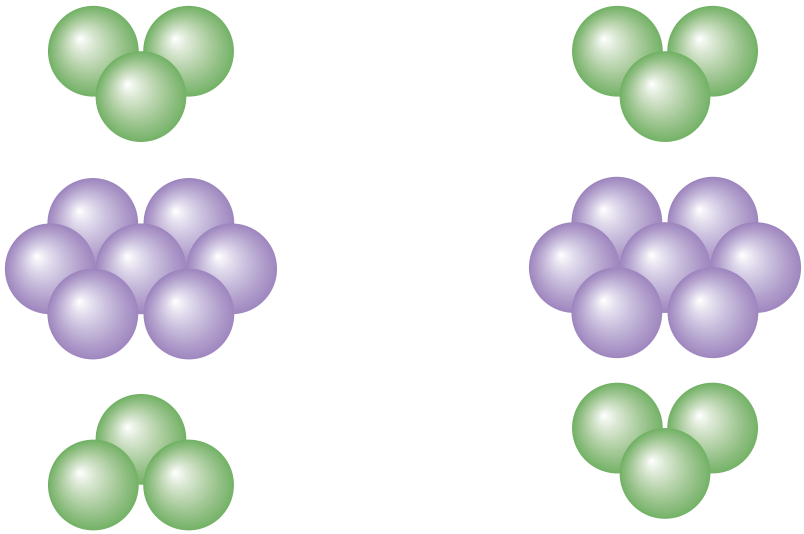


a) Draw the unit cell of sodium.

b) What are the coordination numbers of atoms in sodium and magnesium respectively?

Sodium – 8
Magnesium – 12

c) Which type of packing, (i) or (ii), does magnesium have?
(ii)



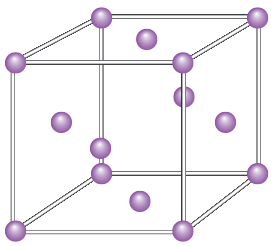
(i) (ii)

2 The face-centred cubic unit cell of iodine is shown below.

There are six molecules at the six faces and eight molecules at the eight corners.

- one molecule at the centre of each of the six faces, giving $6 \times \frac{1}{2}$ molecules;
- One molecule at each of the eight corners, giving $8 \times \frac{1}{8}$ molecules.

$\therefore$ number of molecules per cell $= \left[\left(6 \times \frac{1}{2}\right) + \left(8 \times \frac{1}{8}\right)\right] = 4$



key:
one iodine molecule

Calculate the number of iodine molecules in a unit cell of iodine. Show how you can derive your answer.

51.5 Alloys

Pure metals are widely used. However, pure metals are too weak or too soft for some purposes. They also corrode too easily. Hence the use of pure metals is limited.

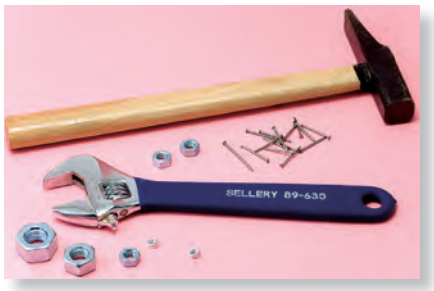


Fig. 51.18 Products made of alloys

By mixing a metal with other metals or non-metals, we can increase the hardness, strength and corrosion resistance of a metal. This process is called alloying and the resulting mixture is called an **alloy**.

You may expect an alloy made of two metals to have properties halfway between the two metals. This may not be true. Sometimes the properties of an alloy are quite different from the pure metals from which it is made. It is these new properties that make alloys so useful.

Compositions of some common alloys

Table 51.2 shows the typical compositions of some common alloys.

N5 Introducing chromium into iron makes iron become corrosion resistant.



(a)



(b)

Fig. 51.19 (a) A five-dollar coin is made of copper and nickel (b) Solder is made of lead and tin

Table 51.2 Typical compositions of some common alloys

Alloy	Approximate composition
Brass	copper 65% zinc 35% CE 2003 Paper 1 Q9(b)(i)
Bronze	copper 90% tin 10%
Mild steel	iron 99.8% carbon 0.2%
Stainless steel	iron 74% chromium 18% N5 nickel 8%
Duralumin	aluminium 95% copper 3% magnesium 1% manganese 1%
Solder	lead 67% tin 33% CE 1999 Paper 1 Q7(a)(iii)(I)
Coinage metal	'silver' coins: copper 75% and nickel 25% 'copper' coins: copper 97%, zinc 2.5% and tin 0.5%
Carat gold*	gold, silver and copper (in various proportions)

* The proportion of gold in its alloy is given by a carat number. Pure gold is 24-carat gold.

alloy 合金　brass 黃銅　bronze 青銅　stainless steel 不銹鋼　duralumin 硬鋁
solder 焊錫

51.6 Comparing the properties of alloys and pure metals

In the following activity, you are going to compare the properties of alloys and those of the pure metals from which they are made.

Activity 51.2

Comparing the properties of alloys and those of the pure metals from which they are made.

We find that each alloy has some properties that are better than the pure metals from which it is made.

1 Appearance — Brass (golden) and coinage metal (silvery or golden) are more attractive in appearance than the pure metals.

2 Hardness — Brass, coinage metal and carat gold are harder than the pure metals.

3 Melting point — Solder has a lower melting point than the pure metals.

4 Corrosion resistance — Brass and solder are more corrosion resistant than the pure metals.



Fig. 51.20 A suspension bridge 'hung' by steel ropes

51.7 Steels

Carbon steels are alloys of iron and carbon. Carbon steels vary in the percentage of carbon they contain. The amount of carbon affects the properties of the steel and its suitability for specific uses.

Mild steels, containing less than 0.2% carbon, are ductile and malleable. Structural steels, containing 0.2 – 0.5% carbon, are harder than mild steels. Tool steels, containing 0.9 – 1.5% carbon, are very hard and strong. Surprisingly, pure iron is extremely soft and is rarely used.

Increasing the amount of carbon can increase the hardness of steel as shown by the graph in Fig 51.21. For example, the steels used to make bobby pins and paper clips contain different amounts of carbon. The steel used for paper clips contains little carbon while that for bobby pins has more carbon and hence is much harder and stronger.

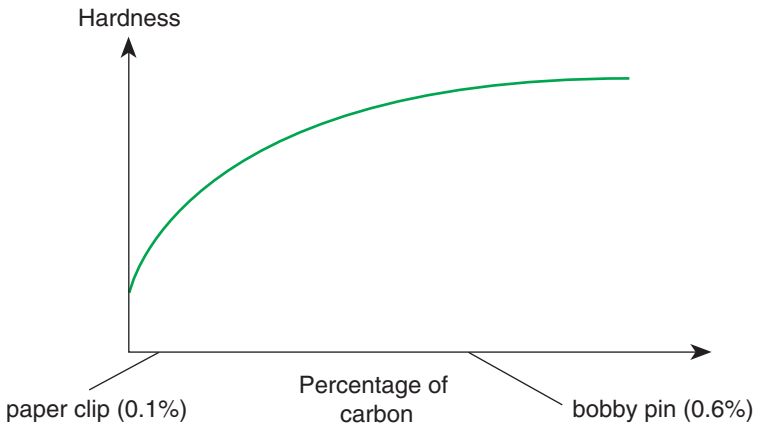


Fig. 51.21 Hardness of steel as a function of percentage of carbon



Fig. 51.22 Bobby pins and paper clips

The addition of small amounts of other metals to steel changes its properties even further, making it suitable for particular purposes. For example, tungsten steel (containing 24% tungsten and 5% chromium) is very hard and very strong. It is used to make high-speed tools for working with metals. Stainless steel (containing 18% chromium and 8% nickel) is four times as strong as mild steel and very corrosion resistant. It is used to make surgical tools, sinks and cutlery.



Fig. 51.23 Products made of stainless steel



Fig. 51.24 Items made of brass

51.8 Brass

Brass is an alloy of copper and zinc. Brass is harder than pure copper and more corrosion resistant. Brass fasteners (screws, bolts, nails and so on) may be used in situations where other metals would rust. Brass also has an attractive finish and hence is used for decorative items such as door knobs and knockers.

Table 51.3 summarizes the uses and properties of some common alloys.

Table 51.3 Uses and properties of some common alloys

	Alloy	Properties	Uses
Iron alloys	mild steel	• ductile • malleable	• nails • chains • cables • machinery
	structural steel	• harder and stronger than pure iron	• girders • vehicles • rails
	tool steel	• very hard • strong	• tool blades • cutlery
	stainless steel	• hard • strong • very corrosion resistant	• surgical tools • sinks • cutlery
Copper alloys	brass (CE 2003 Paper 1 Q9(b)(ii))	• golden appearance • more corrosion resistant than pure copper • harder than pure copper • malleable	• ornaments • buttons • screws • door knobs • taps • pipes
	bronze	• golden appearance • more corrosion resistant than pure copper • resists wearing better than pure copper	• ornaments • statues • church bells • ship's propellers
	coinage metal	• shiny appearance • more corrosion resistant than pure copper • harder than pure copper • malleable	• coins
Tin alloy	solder (CE 1999 Paper 1 Q7(a)(iii)(2), CE 2006 Paper 2 Q9)	• melting point of solder (183 °C) is lower than that of tin (232 °C) and lead (327 °C) • strong	• joining wires and pipes
Aluminium alloy	duralumin	• stronger and more corrosion resistant than pure aluminium • light	• aircraft • spacecraft
Gold alloy	carat gold	• stronger and harder than pure gold	• ornaments • gold coins

CE 2001 Paper 1 Q7(c)(ii)

Checkpoint

1 The ten-dollar coin shown below is made of two alloys.

a)
- The alloy should be soft enough to be minted easily.
- The alloy should be corrosion resistant as coins need to last for a long time.
- The alloy should be wear-resistant as coins are handled frequently.



b) Advantage
Any one of the following:
- More attractive
- Prevent forgery
- Can be distinguished easily from other coins

Disadvantage
Any one of the following:
- Difficult to make
- More expensive
- Possible to detach from the coin

a) What properties should an alloy used for making coins possess? Explain your answer.

b) When compared with using one alloy, using two alloys to make the ten-dollar coin has both advantages and disadvantages. Give ONE advantage and ONE disadvantage of using two alloys to make the coin.

2 The table below shows some properties of six different aluminium alloys.

Alloy	Strength (**** = best, * = worst)	Corrosion resistance	Appearance	Cost
A	*	very good	shiny	low
B	****	fair	dull	high
C	**	fair	dull	low
D	***	very good	shiny	medium
E	***	very good	dull	medium
F	***	fairly good	dull	high

Using the data in the table, choose an alloy for making each of the following articles. Explain each choice. It is more important for students to support their choice with sensible reason(s) rather than giving the right answer.

a) Underwater equipment for drilling oil

b) Window frames for a high-class office

a) D / E is a possible choice because its strength is quite good, very resistant to corrosion and the cost is not too high. Cost is a very important factor for large structures. (B has the best strength but fair corrosion resistance.)

b) D is a possible choice because its strength is quite good, it has very good corrosion resistance, a shiny appearance and the cost is not too high.

51.9 The structure of alloys

Alloys are prepared by melting the components together and cooling the mixture.

If the atoms of the metals being used to form an alloy are about the same size, some of the host metallic atoms will be replaced by the other metallic atoms. This type of alloy is called **substitutional alloy**. Brass is a substitutional alloy in which some of the copper atoms in the host copper metal have been replaced by zinc atoms (Fig. 51.25a).

Sometimes the sizes of the atoms of the components in an alloy differ greatly. The smaller atoms may fit into holes among the closely packed host metallic atoms (Fig. 51.25b). This type of alloy is called **interstitial alloy**. Carbon steel is an interstitial alloy consisting of carbon atoms in the holes among the iron atoms.

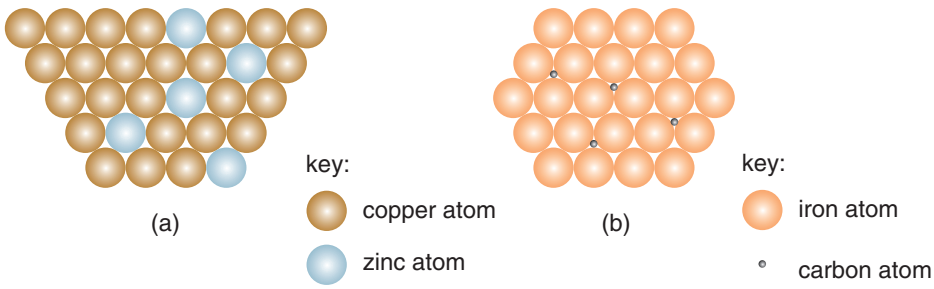


Fig. 51.25 Two types of alloy: (a) a substitutional alloy (e.g. brass) (b) an interstitial alloy (e.g. steel)

51.10 Explaining the properties of alloys in terms of their structures

In a pure metal, all the atoms are of the same size. The layers of atoms can slide past one another easily when a force is applied (Fig. 51.26a). In an alloy, atoms of a different size are added. This distorts the regular structure of the pure metal. The layers of atoms in the alloy are difficult to slide past one another when a force is applied (Fig. 51.26b). Hence the alloy is stronger and harder than the pure metals from which it is made. The alloy is also less malleable and ductile.

CE 1999 Paper 1 Q7(a)(ii)
DSE 2012 Paper 2 Q2(c)(ii)

substitutional alloy 取代合金　　interstitial alloy 間隙合金

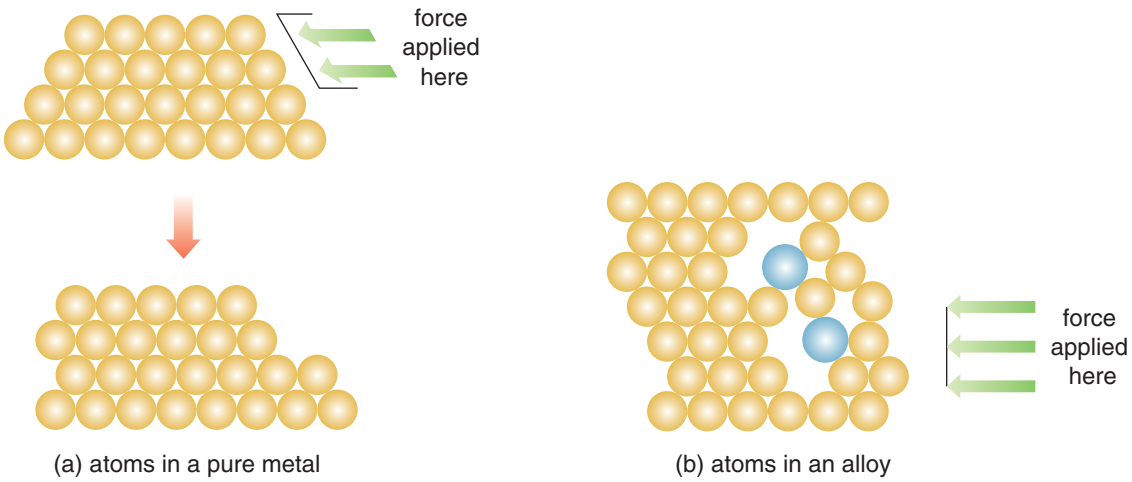


Fig. 51.26 What happens to the atoms in a pure metal and an alloy when a force is applied

The melting point of a solid is very much dependent on its structure. A regularly packed solid has a higher melting point than one with a less regular structure. The structure of an alloy is less regular than those of the pure metals from which it is made. Hence an alloy usually has a lower melting point than the pure metals.

Table 51.4 shows the electrical conductivities of some common metals and alloys.

Table 51.4 Electrical conductivities of some common metals and alloys

Metal / alloy	Electrical conductivity (S m^{-1})
Copper	6.0×10^7
Brass	1.6×10^7
Iron	1.0×10^7
Mild steel	0.6×10^7
Stainless steel	0.2×10^7

Compare the electrical conductivities of copper and brass. The presence of zinc atoms in the copper hinders the movement of mobile electrons. Hence the electrical conductivity of brass is lower than that of copper.

The electrical conductivities of both mild steel and stainless steel are also lower than that of iron.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Shape memory alloys

When a stress is applied to a common metal or an alloy, it will be elastically deformed to less than 1%. Further increase in stress will lead to a permanent change in shape of the metal or alloy. The metal or alloy will remain in that shape even when the stress is released.

Some alloys have a remarkable property. When they are deformed from their original shape, they can revert to their original shape when heated to a certain temperature. They are called shape memory alloys (often called 'memory metals'), a class of metal alloys that 'remember' their original shape.

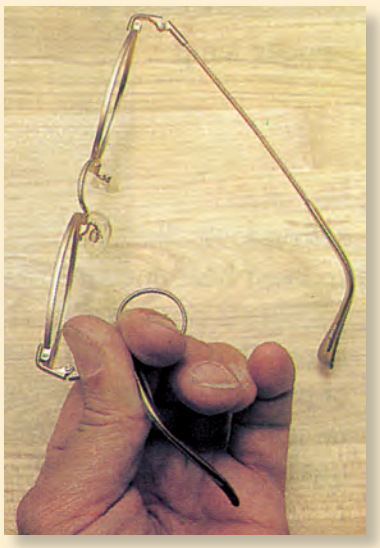


Fig. 51.27 When the frame of the spectacle is bent, it can revert to its original shape when heated to a certain temperature

Different solid phases

Solid, liquid and gas are the three states of matter, but these states can be called by a different name — phases.

A material changes from a solid phase to a liquid phase during melting. Phase changes can take place within a solid also. A solid can have two phases if it has two possible crystalline structures. This ability of changing from one crystalline structure to another gives shape memory alloys their special property.

A common shape memory alloy — Nitinol

A common shape memory material is an alloy of nickel and titanium called *Nitinol*. The alloy is popular because of its high thermal stability, excellent corrosion resistance and high strength.

Nitinol has a crystalline structure called the *austenite* phase. If the austenite phase is cooled under controlled conditions, the material takes on the *martensite* phase.

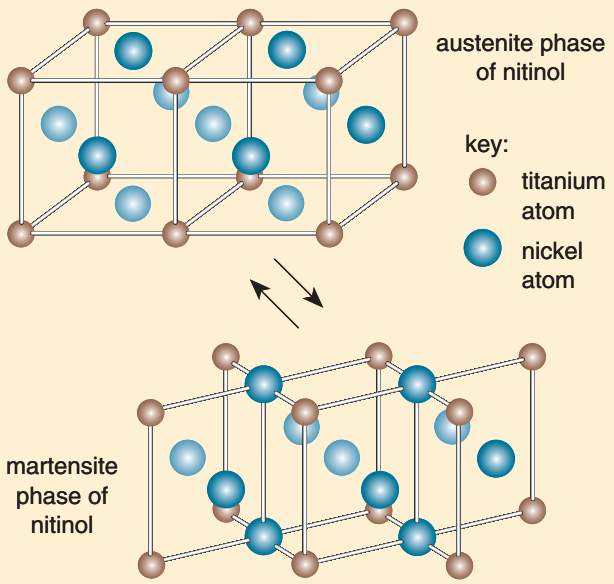


Fig. 51.28 The austenite phase and martensite phase of nitinol

Nitinol 鎳鈦諾　austenite 奧氏體　martensite 馬氏體

The alloy at the martensite phase will change its shape if a stress is applied to it. The alloy will remain in that shape even when the stress is released. However, when the alloy is heated to the austenite phase, the deformation will be recovered and the original shape resumed.

Refer to Fig. 51.29. When nitinol is shaped into a straight wire (a), heated (b) and cooled past its transition temperature (c), it takes on the martensite phase. Notice that its external shape has not changed. It is still straight (d). In its martensite phase, Nitinol is easily bent to a new shape (e). However, when the alloy is heated it recovers its previous shape (f).

Applications of shape memory alloys

Shape memory alloys were originally developed in the United States for the space industry. Now these alloys are used in various areas.

Shape memory alloys are extremely useful for a range of medical applications. *Coronary artery* disease is caused by a plaque in-growth developing on and within an artery's inner wall. This reduces the cross-section of the artery and consequently reduces blood flow to the heart muscle. A *stent* in its 'deformed' shape (having small diameter) contained in a *catheter* can be introduced into an artery. When deployed, the device is triggered by heat from the body and returns to its original 'memorized' shape, providing sufficient force to open the vessel lumen and reinstate blood flow.

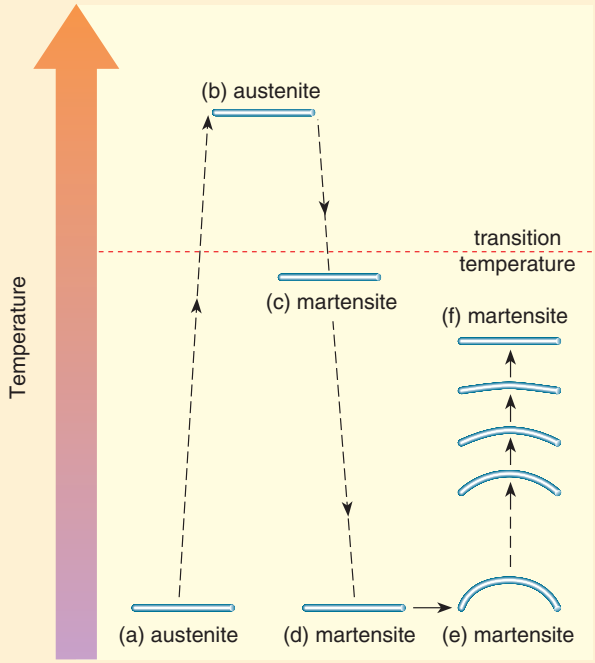


Fig. 51.29 Conversion between austenite and martensite phases as temperature changes

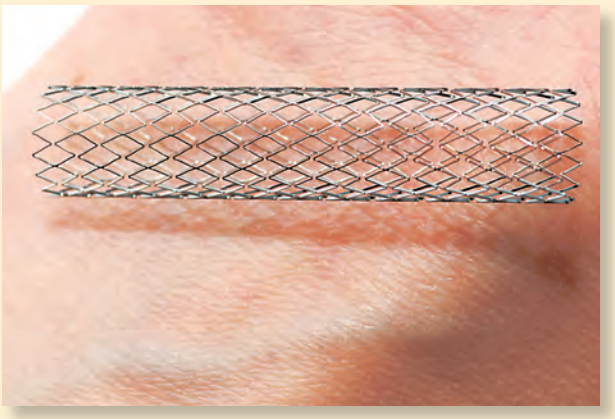


Fig. 51.30 Coronary artery stent

Questions Refer to page T46.

1 Shape memory alloys are used by surgeons in operations to connect bones. Suggest at least TWO other properties that the shape memory alloy should have for this application.

coronary artery 冠狀動脈　　stent 支架　　catheter 導管

2 Suggest another possible use of shape memory alloys.

You may refer to the following websites for some ideas:

- Smart material and micro-machines website of the University of Alberta, Canada
  http://www.cs.ualberta.ca/~database/MEMS/sma_mems/sma.html
- A website demonstrating material science technology by publishing case studies from students at the University of Washington, USA
  http://depts.washington.edu/matseed/mse_resources/Webpage/Memory%20metals/shape_memory_alloys.htm

51.11 Choosing the right material

Why do window designers work with glass, while hockey stick designers do not? Glass allows light to pass through, but it is brittle. If you used it to hit a fast-moving ball, it would shatter.



Fig. 51.31 Hockey sticks are made of wood or composites. These materials are stiff, strong and light

Designers and engineers need to make wise choices when they select the materials for daily commodities, bridges, buildings, etc. The materials must be fit for the job.

As well as performance, there are other factors to consider when choosing the right material:

- Cost — is it affordable?
- Is it durable? How long will it last?
- Environmental impact — what damage might the material do during manufacture, use, or as waste at the end of its use?
- How easily can the material be shaped or joined to other materials?
- Aesthetic appeal — does it look good?

* It is no coincidence that metals, polymers and other materials cluster together on the strength-density chart. These properties reflect the characteristic structure and bonding in each class.

Materials selection charts

Materials selection charts are a graphical way of presenting the properties of materials. Fig. 51.32 is a chart of strength against density. The chart shows that the properties for a given class of materials tend to cluster together*. The chart also highlights key characteristics. For example, polymers are relatively weak compared with metals and composites.

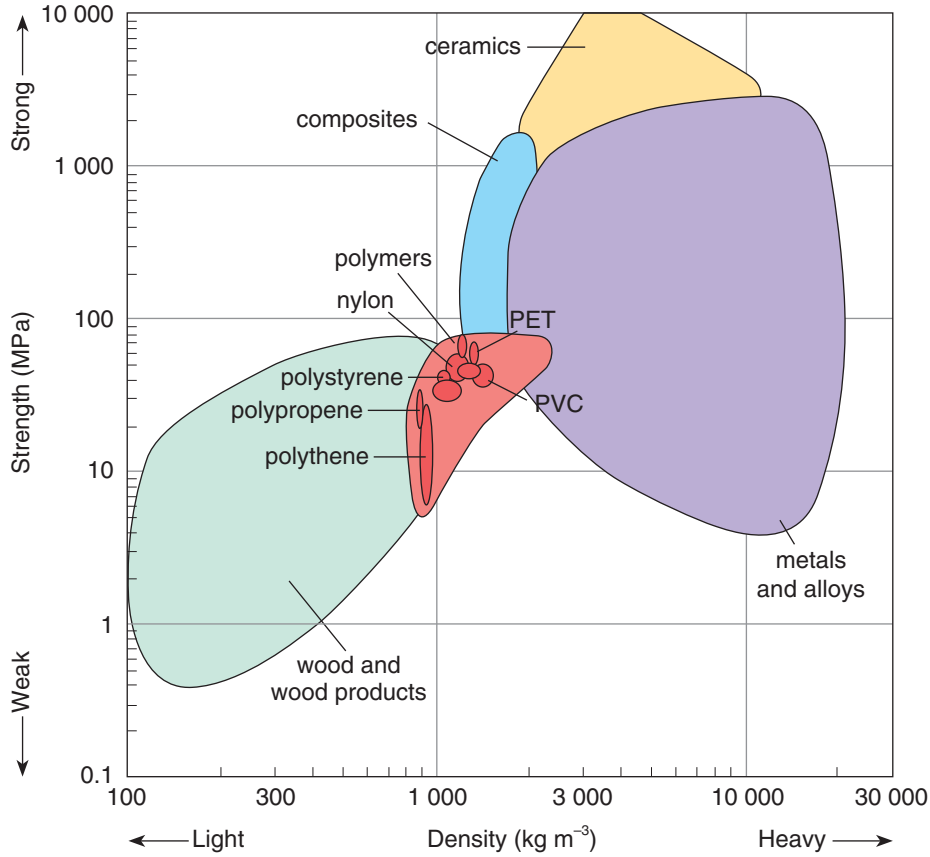


Fig. 51.32 A chart showing the strength against density of some common materials

These two-dimensional charts make it easier to appreciate the range and relative magnitudes of the two properties shown by the plot.

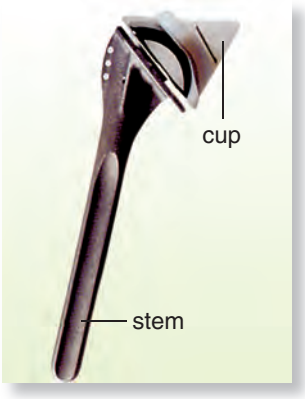
When selecting materials for products, designers have to take into account many competing aims and constraints at the same time: such as light and stiff, strong and cheap, tough and recyclable. Materials selection always involves making trade-offs between competing requirements. Selection charts provide insight into these trade-offs by pairing properties designers must often consider together.

Checkpoint

Checkpoint

a) Any four of the following:
- Strong — resistant to abrasion
- Flexible — would not break when bent
- Corrosion resistant — would not be attacked by body fluids
- Biocompatible — would not be rejected by the body
- High quality at reasonable cost

The diagram shows the two parts of a total hip replacement.



a) State and briefly explain the importance of FOUR necessary properties of materials for making an artificial hip-joint.

b) Suggest why the cup part of the joint is often made of a polymer but the stem is usually made of a metal.
Polymer is smooth, thus the friction is low.
Metal is wear-resistant.

c) An alloy of aluminium and magnesium is used in making lightweight frames. Suggest TWO reasons for not using this alloy for hip-joints.
Too reactive
Not hard / strong enough

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Refer to page T47.

Bags are designed to carry everything from school books to mountaineering gear, or they may simply be a stylish fashion accessory carrying little more than the mass of a lipstick. What governs our choice of material in making these bags?

Fashion handbags	Schoolbags	Mountaineering backpacks
Historically, most designs have been made of leather or other natural materials. Current technologies allow synthetic polymers to replace natural materials. These bags can be made in exotic colours.	A schoolbag must support more mass than a handbag. In the past, the most common materials chosen were natural materials, for example canvas (usually cotton-based).	For mountaineering gear, the material must perform to high standards. Materials for making mountaineering backpacks have evolved from canvas to synthetic polymers.

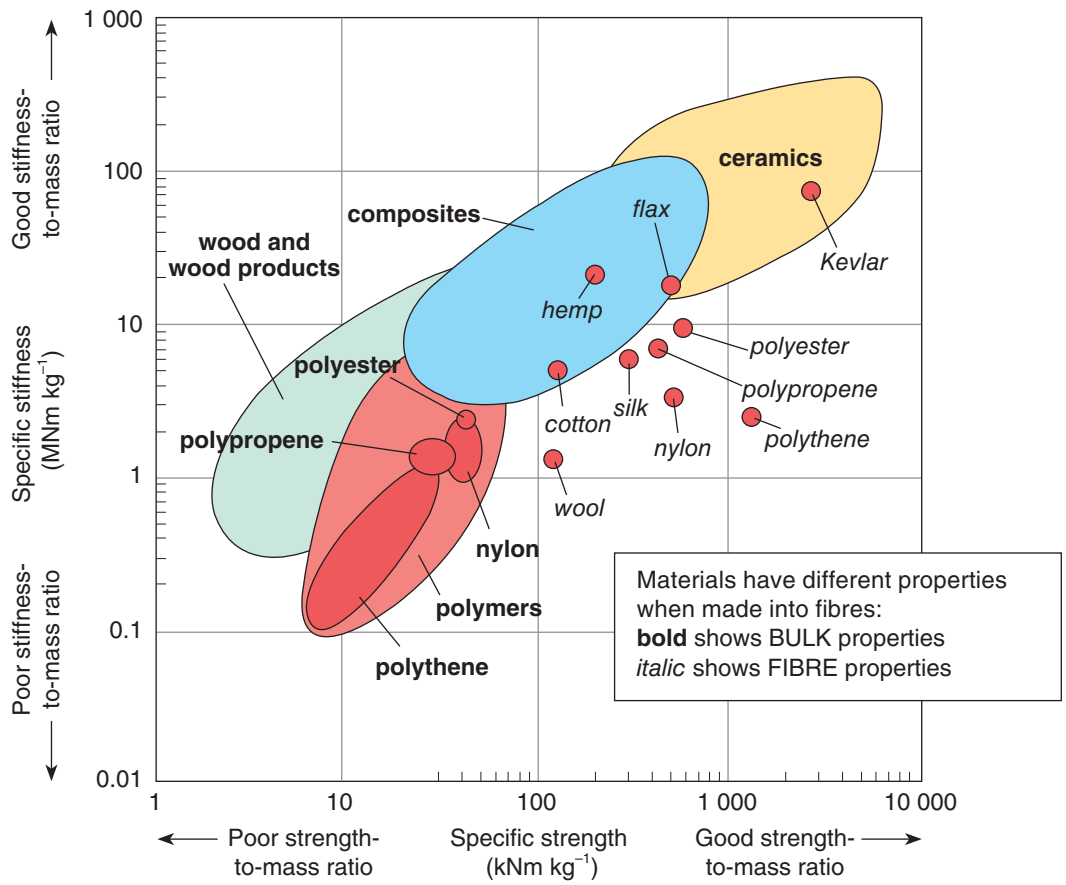
The bags are nearly all made from fabric of some form or another. A fabric is usually made as a flat, flexible sheet by weaving or knitting fibres in bundles. The fibres can be made of natural polymers like silk, cotton or wool, or of synthetic polymers like polythene, nylon or Kevlar.

Suppose you need to choose a fabric for making each type of bags. Each type of bags involves a different set of requirements for the fabric used.

1 The following table shows a list of requirements you may consider when choosing a fabric for each type of bags. Complete the table (using low / medium / high) for the requirements of each type of bags.

Requirement	Fashion handbags	Schoolbags	Mountaineering backpacks
Strength / stiffness			
Mass of bag			
Water resistance			
Tear resistance			
Durability			
Aesthetic appeal			
Allowable cost			

2 For high performance fabrics, we require strength and reasonable stiffness at low mass. The strength of a fabric depends on the type of fibre used. The chart below shows the specific stiffness (i.e. stiffness-to-mass ratio) against specific strength (i.e. strength-to-mass ratio) of some fibres. This allows the comparison of natural and synthetic fibres, and also the comparison of synthetic fibres with the same polymers in bulk form.



Based on the list of requirements you set, choose a fibre for producing the fabric used for making each type of bags mentioned above.

You may refer to the following website for some more materials selection charts:
http://www-materials.eng.cam.ac.uk/mpsite/interactive_charts/default.html

- A chart of strength against density — this chart helps in selecting strong materials that are also lightweight.
- A chart of strength against *toughness** — this chart helps in selecting materials that are not only strong but also tough.
- A chart of strength against cost — this chart helps in selecting strong materials of low cost.

* Toughness measures the ability of a material to withstand *impact loads*.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Car wars

Not so long ago cars were almost exclusively made from iron and steel. But fierce competition, and a push for lighter, more fuel-efficient vehicles has led to the introduction and continued implementation of other materials such as aluminium, plastics and a host of various composites.

Benefits of advanced materials

Plastic composite materials have had the largest impact on how car designers select their materials.

The major advantages of plastics include lightweight, greater design flexibility and parts consolidation potential. These advantages enable carmakers to improve mileage, create unique designs and reduce assembly and total costs.



Fig. 51.33 There is a push for lighter, more fuel-efficient vehicles





Fig. 51.34 The plastic parts of a car — bumper and dashboard

toughness 韌性　　impact load 沖擊荷載

The history of plastics in automotive applications has been a real successful story since the 1980s. The proportion of plastic parts relative to the total vehicle mass has risen continuously since then and currently stands at about 14% (Fig. 51.35).

Industry officials believe that new plastic technology developments will increase the use of plastic composites in exterior body systems and in *chassis* applications.

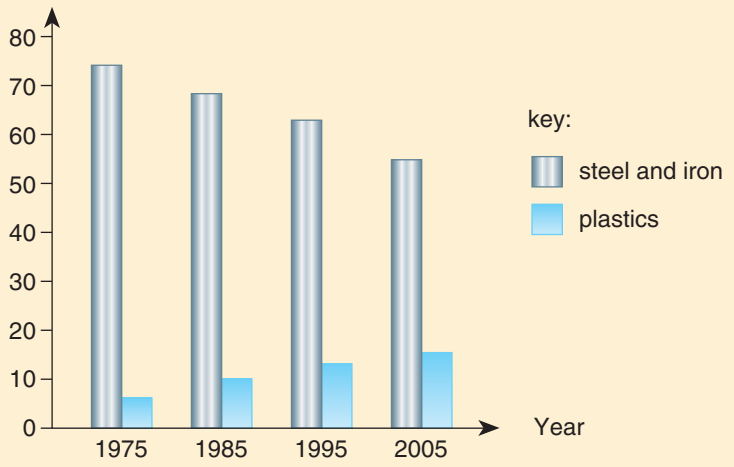


Fig. 51.35 Percentage of material relative to the total vehicle mass

The impact on steel

The steel industry responds to these changes by developing more efficient chassis designs, using higher strength steels and inventing more innovative processing techniques.

Steel producers have worked diligently to create products that are lighter, stronger and easier to manipulate.

Future trends

It is clear that the move towards more dynamic materials will increase. Some analysts predict that firms will start to experiment with metals and alloys, such as titanium and *inconel*, that have become popular in the aerospace industry.

As the producers of different materials fight for their market share, the ideal final result should be an attractive, environmentally-conscious vehicle that is cheap and safe to operate.

Questions Refer to page T50.

In recent decades, more modern materials have replaced those formerly used for making different parts of a vehicle.

1 Give ONE example of an application in which a new material is now employed.

2 What material was formerly used?

3 What newer material is now used?

4 What are the reasons for the change?

chassis 車身底盤　　inconel 英科乃爾（鎳鉻合金）

In the following sections, we will look at some synthetic materials in modern lives, including

- liquid crystals;
- ceramics; and
- nanomaterials.

51.12 What is the liquid crystal phase?

Solid, liquid and gas are the three states of matter, but these states can be called by a different name — phases. A phase is a homogeneous portion of matter separated from other portions of matter by a boundary surface.

The difference between the solid and liquid phases is that particles in the solid phase are highly ordered whereas particles in the liquid phase have no order. However, it turns out that at certain temperatures, many substances exhibit a mesophase* between the solid and liquid phases — the **liquid crystal phase**. The particles in a solid phase have very little transitional freedom. On the other hand, the particles in the liquid crystal phase have some degree of transitional freedom with certain orientation preference in one or two direction(s).

* The Greek word 'mesos' means 'in the middle'.

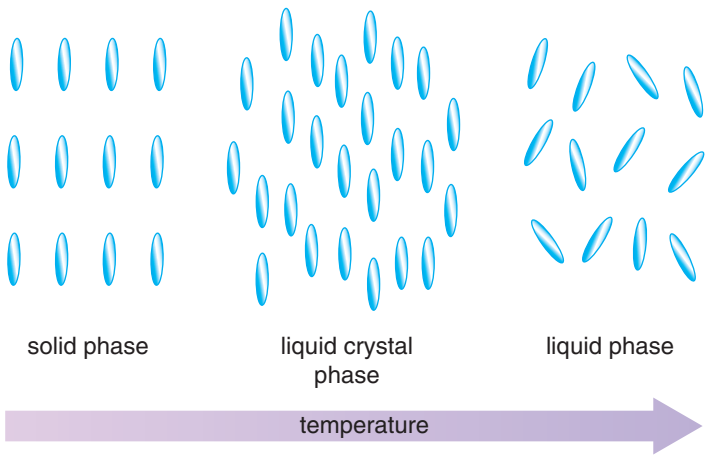


Fig. 51.36 Arrangement of particles in different phases

liquid crystal phase 液晶相

Structural features of substances exhibiting liquid-crystalline behaviour

N6 Liquid crystals would lose the liquid crystal properties at very low temperatures because the compounds solidify.

A very large number of compounds are known to exhibit liquid-crystalline behaviour. Their molecules are often shaped like rods or disks. The most common type of such molecules is rod-shaped (Fig. 51.37). N6

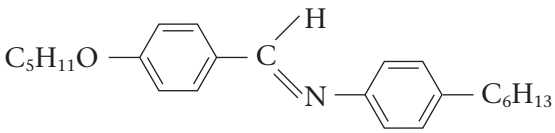


Fig. 51.37 The molecule of a compound that exhibits liquid-crystalline behaviour N7

N7 Many liquid-crystalline materials contain
- benzene rings;
- polar groups.

DSE 2014 Paper 2 Q2(a)(ii)

These molecules have an uneven distribution of electrons. This leads to intermolecular forces which, over a certain range of temperatures, are strong enough to cause the molecules to line up in the same overall direction but not strong enough to hold them firmly in one place.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

The discovery of liquid crystals

The discovery of liquid crystals is usually attributed to an Austrian botanist, Friedrich Reinitzer.

In 1888, he was trying to measure the melting point of *cholesteryl benzoate* and found that it seemed to have two melting points. At 145.5 °C the crystals formed a cloudy liquid and at 178.5 °C the liquid became clear. Upon cooling, the reverse process occurred.

cholesteryl benzoate crystals $\underset{}{\overset{\text{first melting point}}{\rightleftharpoons}}$ a cloudy liquid $\overset{\text{second melting point}}{\rightleftharpoons}$ a clear liquid

A colleague of Reinitzer, Otto Lehmann, examined the in-between phase of cholesteryl benzoate with a polarized light microscope. The sample being examined was sandwiched between a pair of crossed polarizers. He was surprised to find that it had different *refractive indices* in different directions. This is a property normally associated with ordered crystalline solids. Lehmann was surprised to find such a property shown by a liquid. The observation implied that the molecules in the liquid were arranged in an orderly fashion. This was the origin of the term 'liquid crystal'.

cholesteryl benzoate 膽甾醇苯酸酯　　refractive index 折射率

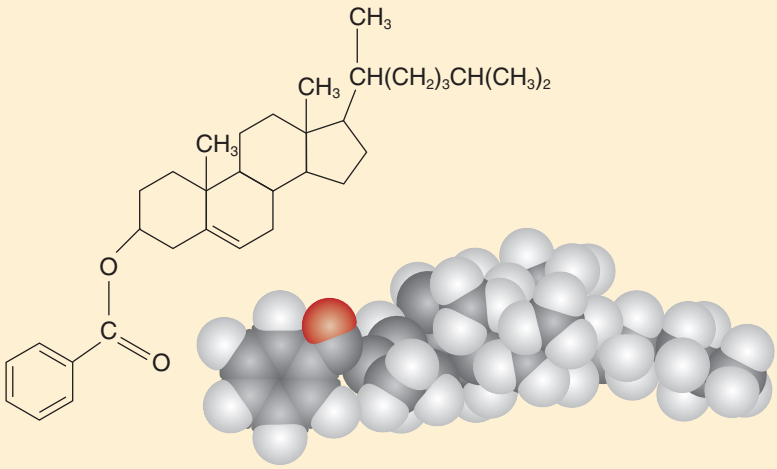


Fig. 51.38 The rod-like molecular structure of cholesteryl benzoate

Lehmann and Reinitzer were engaged in basic research, and neither dreamed of an application for their findings. But it was not until the 1960s that investigations began in both basic research and practical uses for liquid crystals. By the early 1970s consumer products such as watches and calculators containing small liquid crystal displays flooded the market. Today liquid crystal displays are among the most popular forms of electronic information displays.



Fig. 51.39 A computer screen that uses liquid crystals for display

Question

Identify as many daily products using liquid crystal display as possible.

Some examples students may suggest:

- LCD digital display pressure gauges
- Wall thermostat with LCD
- LCD digital clock
- Mobile phone
- LCD for lift / elevator
- Touch screen LCD
- On camera LCD monitor
- Alcohol breath tester with LCD

51.13 Categories of liquid crystal phases

Depending on the nature of the molecular ordering, liquid crystal phases can be classified into three categories: **nematic**, **smectic** and **cholesteric**.

Fig. 51.40 shows the ordering of molecules in the nematic and smectic phases. Molecules in the nematic phase tend to point in the same direction. This preferred direction is called the **director**. Molecules in the smectic phase show some degree of two-dimensional order. They align towards the direction of the director and tend to arrange in layers.

DSE 2013 Paper 2 Q2(c)(i)

DSE 2015 Paper 2 Q2(c)(i)

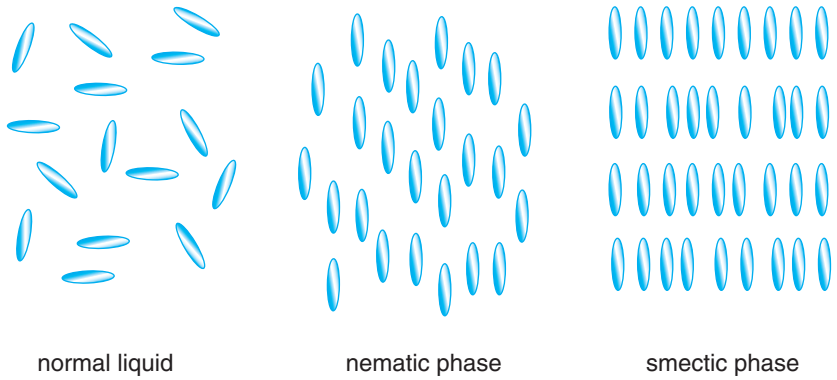


Fig. 51.40 Diagrams showing the ordering of molecules in different phases

DSE 2013 Paper 2 Q2(c)(ii)

Cholesteric or chiral nematic phase arises when the molecules that make up the nematic phase are chiral. In this phase, the molecules line up in positions that they are slightly twisted from the molecules next to them. We can regard the overall arrangement of the molecules as a stack of very thin layers of nematic molecules with the director in each layer twisted with respect to those above and below (Fig. 51.41).

N8 Refer to page T50.

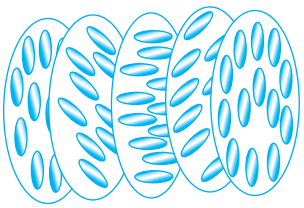


Fig. 51.41 A stack of very thin layers of nematic molecules with the director in each layer twisted

51.14 Operation of a typical liquid crystal display

The first really practical liquid crystal display (which is still used today) was developed in 1970.

Fig. 51.42 shows the structure of a typical liquid crystal display. A thin layer of liquid crystal is sandwiched between two plates of glass coated with indium tin oxide for conducting electricity.

nematic 向列的　smectic 近晶的　cholesteric 螺旋型的　director 長軸方向

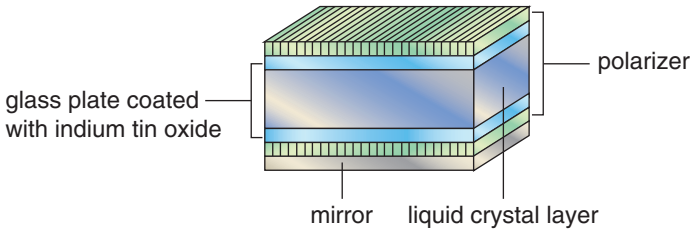


Fig. 51.42 Structure of a typical liquid crystal display

Light passes through a polarizer, enters the sandwich and then leaves it by passing through a second polarizer at right angles to the first.

Fig. 51.43 shows the representation of a typical liquid crystal display. The upper glass has a chemical treatment which makes the liquid crystal molecules close to it line up in a particular direction. The lower glass plate has been treated so that the liquid crystal molecules near it line up at right angles to the first direction. The intervening molecules gradually rotate their relative orientation to accommodate the 90° change from one glass plate to the other. The polarized light will be guided by the molecules. Hence the plane of polarization of the light passing through the liquid crystal layer will be rotated through 90°. As a result, the polarized light can pass through the second polarizer, strike the mirror and reflect back. We can see a white background. This is the 'off' state of the device. N9

N9 Liquid crystal displays are relatively inefficient in terms of power use. Liquid crystals act as optical filters to filter off the light from the backlight to give dark spots. Only a small fraction of light passes through them and makes up the images.

Reference Website
Refer to the following website for an animation showing the technology used in a LCD monitor:
http://www.youtube.com/watch?v=O3aITfU_UvE
(accessed May 2015)

Animation
Liquid crystal display

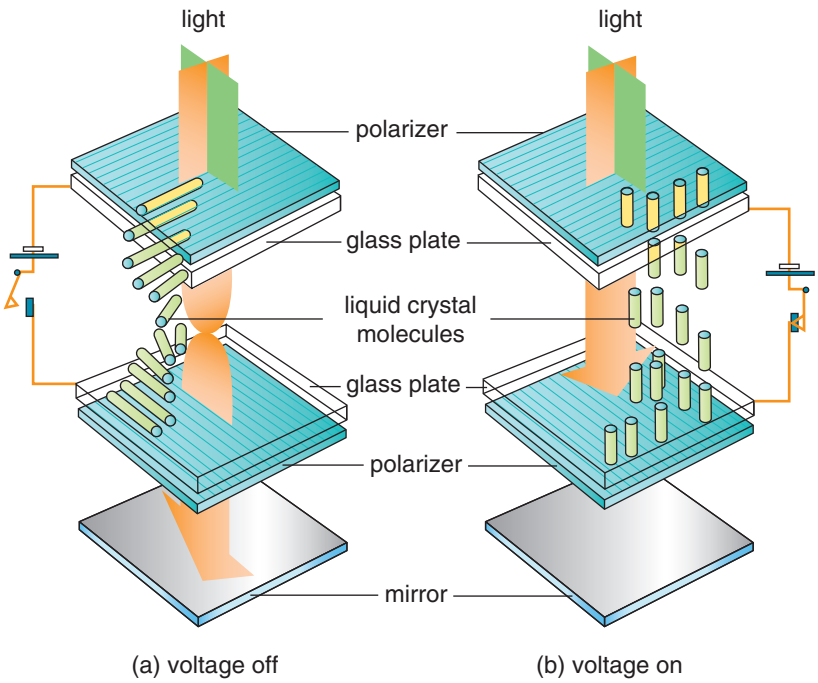


Fig. 51.43 Explaining the operation of a typical liquid crystal display
(a) The 'off' state (b) The 'on' state



Fig. 51.44 A quartz watch with a liquid crystal display

N10 Refer to page T51.

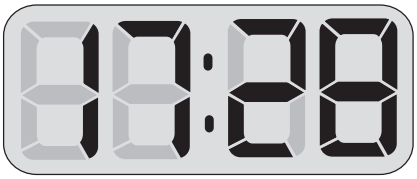


Fig. 51.45 The 7-segment sessions for the numerical display of a watch

* Notice that small and inexpensive liquid crystal displays are often reflective, which means to display anything, they must reflect light from an external light source.



Fig. 51.46 Using a liquid crystal thermometer to read body temperature

When a voltage is applied across the electrodes, it also applies an electric field across the liquid crystal. The liquid crystal molecules line up parallel to the field and no longer rotate the plane of polarization of the light (Fig. 51.43b). In this 'on' state, light cannot pass through the second polarizer to reflect off the mirror. The area appears dark. N10

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Liquid crystal displays in calculators or watches

The most basic 7-segment liquid crystal numerical display is widely used in watches, clocks and similar items.

The liquid crystal layer is sandwiched between two glass plates that feature seven individually chargeable electrodes. These seven pairs of electrodes are arranged in the shape of an '8'.

Fig. 51.45 shows the 7-segment sessions for the numerical display of a watch. When there is no electric field, the light entered can pass through both polarizers and be reflected by the mirror beneath. This particular segment therefore appears as clear (colourless).

However, when an electric field is applied across the electrodes, light cannot pass through the second polarizer and thus no light is reflected by the mirror. This causes that segment to appear as a dark region. By manipulating these seven segments of electrodes, numbers 0 to 9 can be displayed accordingly*.

The main advantage of a liquid crystal display is low energy consumption. More advanced liquid crystal displays need back light, colour filters and advanced electronics to display complex figures.

Liquid crystal thermometers

The colour of light reflected by cholesteric liquid crystals is temperature-dependent. Therefore liquid crystals can serve as thermometers. A liquid crystal thermometer contains cholesteric liquid crystals in a plastic strip that changes colour to indicate different temperatures.

A liquid crystal thermometer attached to the skin can measure temperature variations of the skin. This can be useful in the detection of skin cancer, as tumors have different temperatures than surrounding tissues.

Extension

51.15 Ceramics

Ceramics are solid inorganic compounds that are formed by the application of heat. A ceramic comprises at least two elements, one element is a non-metal (e.g. carbon, silicon, etc.), the other element(s) may be metal(s) or non-metal(s).

For example, magnesia* (MgO) is a ceramic since it is a solid compound of a metal bonded to the non-metal oxygen. Silicon carbide is also a ceramic because it is a solid compound of two non-metals.

Ceramics can be divided into two classes: traditional and advanced. Traditional ceramics include clay products such as porcelain, silicate glass and cement.

Advanced ceramics include oxide ceramics such as alumina (Al_2O_3), and non-oxide ceramics such as silicon carbide (SiC) and silicon nitride (Si_3N_4). These materials have high-tech engineering, electronic and biomedical applications. Table 51.5 shows how their properties compare with those of aluminium and carbon steel.

* The addition of the letter 'a' to the end of the name of an element implies that one is referring to the oxide of that element. For example, alumina is the oxide of aluminium.

Table 51.5 Comparing the properties of advanced ceramics with those of aluminium and carbon steel

Material	Melting point (°C)	Density (g cm^{-3})	Hardness (Mohs)*	Thermal conductivity (W m^{-1} K^{-1})	Electrical conductivity (S m^{-1})
Alumina (Al_2O_3)	2 054	3.98	9	39	$< 10^{-13}$
Silicon carbide (SiC)	2 830	3.20	9	71	$1 - 10^{-9}$
Silicon nitride (Si_3N_4)	1 900	3.30	9	33	$< 10^{-12}$
Aluminium	660	2.70	3	237	37×10^{6}
Carbon steel	1 515	7.85	5	51.9	6.25×10^{6}

* Numbers on the Mohs hardness scale range from 1 for talc, a very soft material, to 10 for diamond, the hardest known substance.

Ceramics have high melting points, and high thermal and electrical insulating properties. They are harder and more resistant to wear and corrosion than metals. Moreover, they maintain much of their strength at high temperatures, whereas metals either melt or corrode because of oxidation. Unfortunately, ceramics are brittle, as anyone who has dropped a coffee cup knows.

Extension

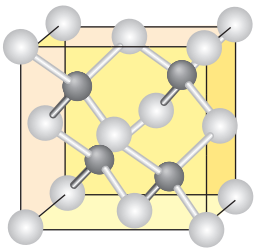


Fig. 51.47 One unit cell of the cubic form of silicon carbide. The silicon atoms are located at the corners and face centres, while the carbon atoms occupy the cavities between four silicon atoms

In general, most ceramics are:

- hard;
- wear-resistant;
- brittle;
- *refractory*;
- thermal insulators;
- electrical insulators;
- nonmagnetic;
- oxidation resistant; and
- chemically stable.

51.16 The structure of silicon carbide

Silicon carbide (SiC) is a covalent ceramic. The building block for silicon carbide is the tetrahedron SiC_4.

Silicon carbide exists in many different crystalline forms, the simplest of which is the cubic form (Fig. 51.47). In the giant network structure, each silicon atom is bonded tetrahedrally to four carbon atoms, and each carbon atom is bonded tetrahedrally to four silicon atoms. As a result of its giant network structure, silicon carbide is hard, has a high melting point, and shows electrical and thermal insulating properties.



51.2 Explaining the brittleness of silicon nitride

Silicon nitride (Si_3N_4) is a covalent network solid in which each Si atom is bonded to four N atoms and each N atom is bonded to three Si atoms.

Explain why silicon nitride is more brittle than a metal such as copper.

Solution

The brittleness of ceramics is due to the strong chemical bonding. In silicon nitride, the strong covalent bonds prevent the layers of atoms from sliding past each other when a force is applied to the solid. As a result, the solid cannot deform to relieve the stress. It maintains its shape up to a point, but then the bonds give way suddenly and the material fails.

By contrast, when a force is applied to a metal, the layers of atoms can slide past each other easily.

refractory 耐火

Extension

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Why are oxide ceramics more corrosion resistant than metals?
Oxide ceramics are oxidation resistant and chemically stable.

51.17 Applications of ceramics

Refractories

Animation
Piezoelectricity in ceramics structure

Metal smelting requires the use of refractories for lining furnaces and keeping the heat within the furnace hearth. Ceramics have high melting temperatures and thermal insulating property. Refractories are generally made of ceramics.

Mechanical applications

Ceramics are hard, wear-resistant and corrosion resistant. They are used to make cutting tools, valves and bearings.

Due to the refractoriness of ceramics and their ability to sustain high loads at high temperatures, ceramic engines are used for transportation and ceramic turbines are used for electricity production.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

In a catalytic converter, the catalysts are coated onto a ceramic honeycomb structure. Discuss why ceramic is used in this case.

All catalytic converters function on essentially the same principal. The inside of some converters can reach over 1 000 °C under normal operation. Thus, the converter housing is designed to reach extremely high heat.
Ceramic is used in catalytic converters because it is both heat resistant and corrosion resistant.

51.18 Nanomaterials

A nanomaterial refers to any organic or inorganic material that has an average building block or particle size between 1 – 100 nanometres.

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One nanometre (nm) is equal to one-billionth of a metre, 10^{-9} m. Figure 51.48 shows the nanometre in context. Atoms are below a nanometre in size, whereas many molecules range from a nanometre upwards.

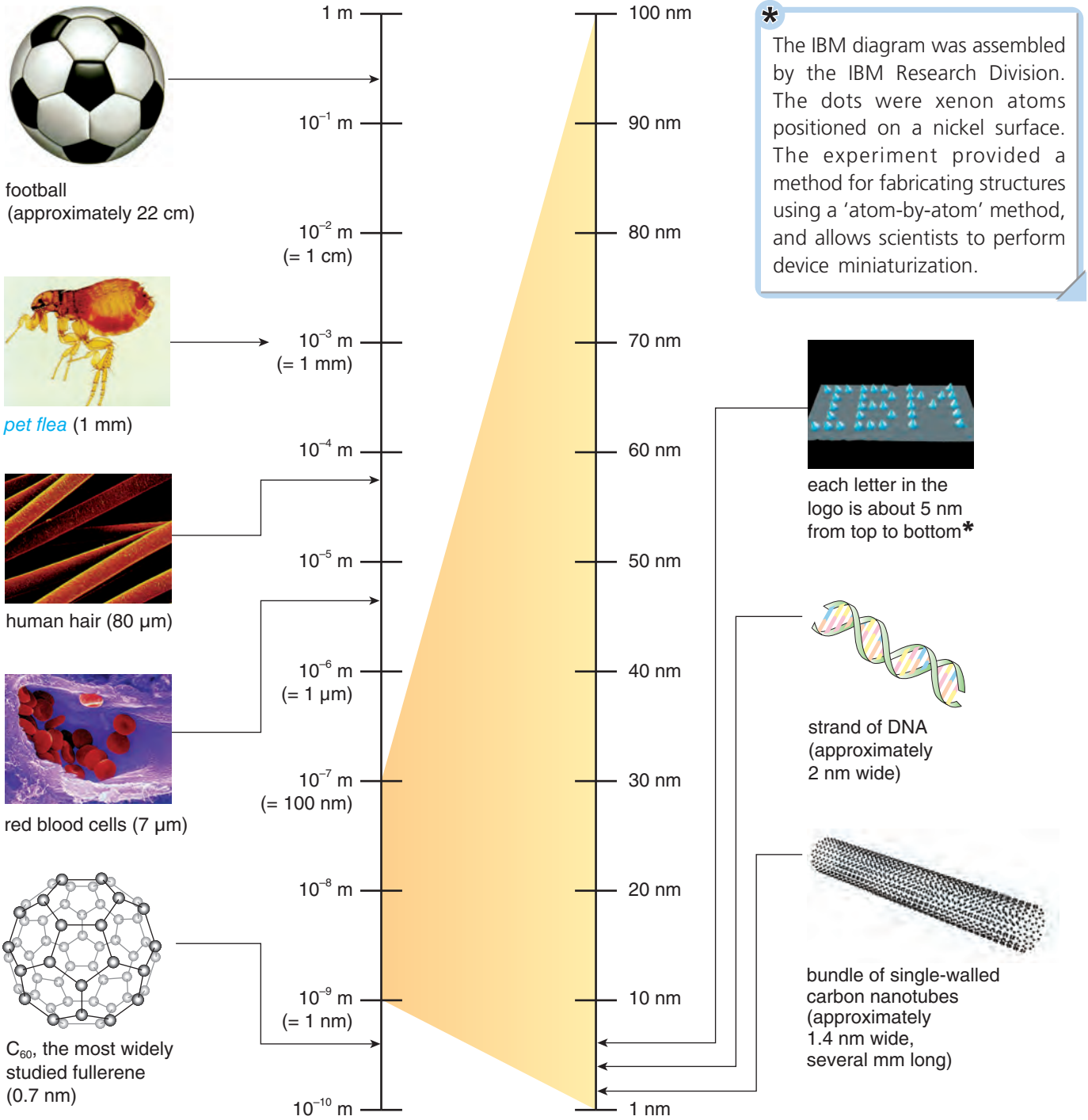


Fig. 51.48 A length scale showing the nanometre in context. The length scale on the left ranges from 1 m to 10^{-10} m. The section from 10^{-7} m (100 nm) to 10^{-9} m (1 nm) is expanded on the right. Atomic scale is at about 0.1 to 0.2 nm

pet flea 跳蚤

The size range that holds so much interest is typically from 100 nm down to the atomic level (approximately 0.2 nm), because it is in this range that materials can have different or enhanced properties compared with the same materials at a larger size. A main reason for this change in behaviour is an increase in relative surface area.

As a particle decreases in size, a greater proportion of the atoms are found at the surface compared to those inside*. Hence a nanoparticle has a much greater surface area per unit mass compared with a larger particle.

* For example, a particle of size 30 nm has 5% of its atoms on its surface. A particle of size 10 nm has 20% of its atoms on its surface. A particle of size 3 nm has 50% of its atoms on its surface.

Categories of nanomaterials

We may categorize nanomaterials in the following way:

- one-dimensional nanomaterials such as thin films or surface coatings that have one dimension in the nanoscale;
- two-dimensional nanomaterials such as nanowires and nanotubes that have two dimensions in the nanoscale;
- materials that have three dimensions in the nanoscale such as nanoparticles and buckminsterfullerene (C_{60}).

51.19 Applications of nanomaterials

Invisible sunscreens

Most sunscreen lotions are oil-in-water emulsions*. Zinc oxide is used in some sunscreens as it strongly absorbs ultraviolet light which causes sunburn and damage to the skin. These sunscreens appear white as zinc oxide reflects the visible light.

* An oil-in-water emulsion consists of tiny droplets of oil suspended in water.

Nanoparticles of zinc oxide are smaller than the wavelength of visible light, they no longer reflect the visible light*. Hence lotions containing nanoparticles of zinc oxide are transparent*, but they still absorb ultraviolet light.

* The nanoparticles do not reflect the visible light because the light is able to pass between the particles.



Fig. 51.49 People having a sunbath under sunlight usually use sunscreens

* Transparent sunscreens are more appealing to consumers. However, the use of nanoparticles in sunscreens and cosmetics has raised a number of concerns about consumer safety. N11

N11 Refer to page T53.

Catalysts

Many catalysts work by providing a surface on which the reactant particles can bond. One method to increase the activity of such catalysts is to increase their surface area so that there are more space for reactant molecules to bond.

Nanoparticles have a much greater surface area per unit mass than particles of larger sizes. Hence nanoscale catalysts provide a higher catalytic activity. They open the way for numerous innovations to make chemical processes more efficient and resource-saving.

Reference Website

- Refer to the following website for a video clip on a nanotechnology-enabled battery:
  http://www.nanowerk.com/nanotechnology/videos/Smart_nanobattery.php
  (accessed May 2015)

Medicine N12

Nanotechnology helps medicine with devices in the treatment and diagnosis of diseases. For example, *pancreatic* cells may be placed inside microcapsules with nanoscale pores. The pores in the capsules are big enough to allow insulin to flow out of the capsules but small enough to prevent the entry of harmful antibodies. In this way, diabetic patients can obtain insulin without injecting it.

N12 Refer to page T53.

Worksheet **Assessment Form**
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.



Internet Search & Presentation

Discovery and applications of nanomaterials

Task

When scientists want to share their findings and proposals with other scientists, they often do it at international meetings.

Your teacher will divide the class in groups. Each group should assume the role of a prominent nanoscientist working on a nanomaterial application in an international meeting. Give a 4-minute presentation on one application listed below or you can choose another application with the teacher's consent. Support your presentation with various aids, such as timelines, diagrams, etc.

Hints for the search Refer to page T54.

1 Give a brief history about the development of nanotechnology.

nanotechnology 納米科技　　pancreatic 胰臟的

2 Discuss one application of nanomaterials. Some applications are listed below.

a) Paint that resists chipping

Nanopaints are more resistant to chipping and have richer and brighter colours than traditional paints. The nanopaints contain tiny ceramic particles added to a liquid clearcoat. Links between the particles can create a very dense and smoothly structured network that provides a protective layer.

b) High density storage media

New nanomedia could have a storage density that is a million times higher than that of current CDs and DVDs.

c) Drug delivery systems

Nanotubes could serve as drug delivery systems. Researchers have attached fluorescent markers and proteins to nanotubes and mixed them with living cells. They discover that (from the fluorescent markers) the nanotubes enter the cells and 'deliver' the proteins to the cells. The nanotubes do not seem toxic to the cells, but lots more research are needed.

d) Clean energy

Future cars may use non-polluting hydrogen fuel cells. Today, hydrogen fuel is expensive to make. However, with catalysts made from nanoclusters, it may be possible to generate hydrogen from water by photocatalytic reactions. Novel hydrogen storage systems could be based on carbon nanotubes and other lightweight nanomaterials.

Reference websites

1 You may start your search from the following websites as appropriate:

A brief history about the development of nanotechnology
- University of Reading — Peter Harris' site on nanotube
  http://www.personal.reading.ac.uk/~scsharip/tubes.htm

Paint that resists chipping
- Mercedes-Benz Nanopaint — article on benefits, material, and paint process
  http://www.auto123.com/en/news/mercedes-benz-nano-paint/46895/

High density storage media
- An article on nanoscale memory
  http://uw.physics.wisc.edu/~himpsel/memory.html

Drug delivery systems

- An article on linking up amino groups with carbon nanotubes
  http://nanotechweb.org/articles/news/2/3/1/1

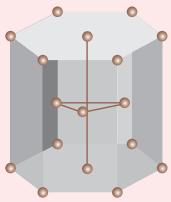
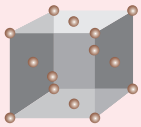
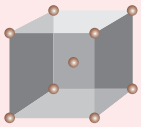
2 Extend your search using search engines and appropriate keywords.

- http://www.zoo.com/Zoo-Site/
- http://www.google.com
- http://www.yahoo.com
- http://www.searchalta.com/

Summary

1 In metals the particles are arranged in a very regular way, giving rise to crystals.

2 The following table summarizes the information about the different types of metallic crystal structures.

	Hexagonal close-packed (h.c.p.)	**Cubic closed-packed (c.c.p.)**	**Body-centred cubic (b.c.c.)**
Stacking pattern	ababab	abcabcabc	odd-numbered layers of atoms are identical; even-numbered layers of atoms are identical
Unit cell			
Number of atoms per unit cell	6	4	2
Coordination number	12	12	8
Examples	magnesium, zinc	aluminium, copper, lead, silver	alkali metals, iron, manganese

3 An alloy is made by mixing a pure metal with other metals or non-metals.

4 Alloys have properties that are better than the pure metals from which they are made, including

a) appearance;

b) hardness;

c) melting point; and

d) corrosion resistance.

5 a) At certain temperatures, many substances exhibit a mesophase between the crystal and liquid phases — the liquid crystal phase.

b) Compounds that exhibit liquid-crystalline behaviour often have rod-shaped or disk-shaped molecules.

c) A common practical use of liquid crystals is liquid crystal display.

Extension

6 a) Ceramics are solid inorganic compounds that are formed by the application of heat. A ceramic comprises at least two elements, one of which is a non-metal.

b) Ceramics have giant network structures and hence they have high melting points, high thermal and electrical insulating properties. They are harder and more resistant to wear and corrosion than metals.

7 a) A nanomaterial refers to any organic or inorganic material that has an average building block or particle size between 1 – 100 nm (1 nm = 10^{-9} m).

b) Nanomaterials can be used in invisible sunscreens, catalysts and medicine.

Exercise

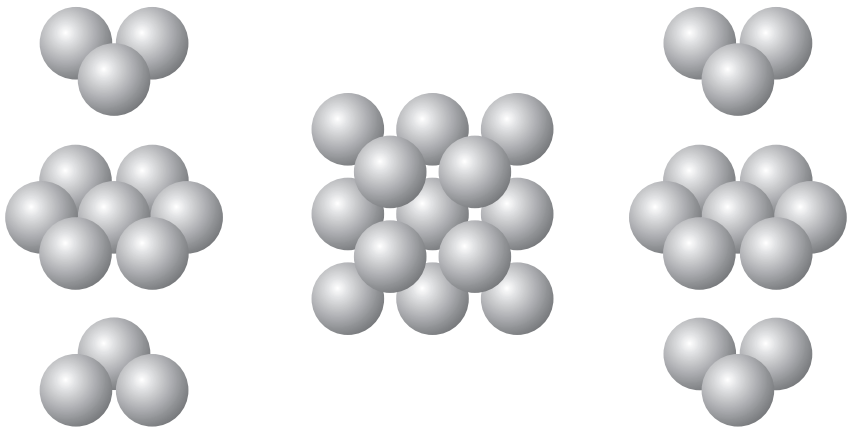
Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following table on information for the three structures of metals.

Structure	Stacking pattern	Coordination number	Example
Body-centred cubic	odd-numbered layers of atoms are identical; even-numbered layers of atoms are identical	8	alkali metals, iron, manganese
Cubic close-packed	abcabcabc	12	aluminium, copper, lead, silver
Hexagonal close-packed	ababab	12	magnesium, zinc

2 The spheres in the following diagrams show three different types of structures.



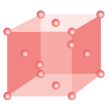
(i) Cubic close-packed structure

(ii) Body-centred cubic structure

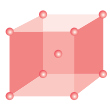
(iii) Hexagonal close-packed structure

a) Name each type of structure represented by the spheres.

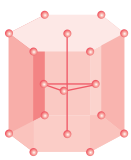
b) Draw a unit cell for each type of structure.



Cubic close-packed structure



Body-centred cubic structure



Hexagonal close-packed structure

3 Complete the following table which lists the constituents of some common alloys.

Alloy	Constituents
Brass	copper, zinc
Bronze	copper, tin
Mild steel	iron, carbon
Stainless steel	iron, chromium, nickel
Duralumin	aluminium, copper, magnesium, manganese
Solder	lead, tin
Coinage metal	copper, nickel
Carat gold	gold, silver, copper

Extension

4 a) Give FIVE general properties of ceramics.

a) Any five of the following:
- Hard
- Wear-resistant
- Brittle
- Refractory
- Thermal insulators
- Electrical insulators
- Nonmagnetic
- Oxidation resistant
- Chemically stable

b) Give TWO examples of oxide ceramics.
Alumina (Al_2O_3) Magnesia (MgO)

c) Give TWO examples of non-oxide ceramics.
Silicon carbide (SiC) Silicon nitride (Si_3N_4)

5 List FIVE factors you would consider when choosing a material for a particular job.
- Cost
- Durability
- Environmental impact
- How easily can the material be shaped or joined to other materials?
- Aesthetic appeal

Part II Structured questions

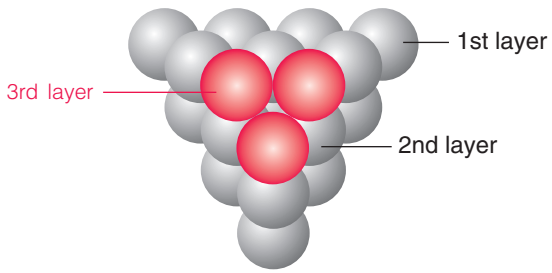
Answers for the questions of the public examinations in Hong Kong are not provided.

6 The arrangement of atoms in metals can be described by the close-packing of spheres.

a) Which close-packed structure does the 'ababab' stacking pattern describe?
Hexagonal close-packed structure

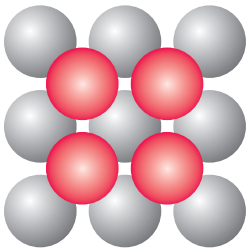
b) Which close-packed structure does the 'abcabc' stacking pattern describe?
Cubic close-packed structure

c) The following diagram shows the first and second layers of close-packing spheres.



Draw a third layer of spheres on the diagram to show the 'abcabc' stacking pattern.

7 Some metals have a body-centred cubic structure. The following diagram shows the first layer of atoms in the structure.



Draw a second layer of atoms on the first layer to show a body-centred cubic structure.

8 The table gives some data of three metals.

Element	Crystal structure
Copper	cubic close-packed (c.c.p.)
Sodium	body-centred cubic (b.c.c.)
Zinc	hexagonal close-packed (h.c.p.)

a) What is meant by the term 'coordination number'?
The coordination number is defined as the number of atoms (or ions) immediately surrounding an atom (or ion) in a crystal lattice.

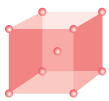
b) What is the coordination number of an atom in

i) a cubic close-packed structure?
12

ii) a body-centred cubic structure?
8

c) Draw the unit cell of a sodium structure and use it to explain why this unit cell contains 2 sodium atoms.

c)



There is one atom in the centre of the body of the unit cell plus eight atoms at the corners of the cube.

$\therefore$ number of atoms per cell = $1 + \left(8 \times \frac{1}{8}\right) = 2$

d) Titanium transforms from the b.c.c. structure to the h.c.p. structure when it cools. Suggest, giving reasons, whether titanium contracts, expands or remains unchanged in volume when this cooling takes place.
Atoms are more closely packed in a hexagonal close-packed structure than in a body-centred cubic structure.
Thus, titanium contracts in addition to normal thermal contraction.

9 a) Name TWO metals with a close-packed structure and TWO metals with a body-centred cubic structure.

b) Suppose one particular metal can have either a close-packed or a body-centred cubic structure. In which of these two forms would you expect it to have

i) a higher density;
Close-packed structure
Atoms are closer 'on average'.

ii) a higher melting point?
Close-packed structure
Atoms are closer and hence more strongly bonded.

Explain your answer in each case.

a) Metals with a close-packed structure
Any two of the following:
- Aluminium
- Copper
- Lead
- Magnesium
- Silver
- Zinc

Metals with a body-centred cubic structure
Any two of the following:
- Alkali metals
- Iron
- Manganese

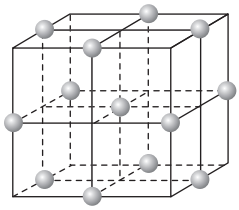
10 There are several crystalline forms of metallic iron. Below 906 °C, iron exists in the α-form, which has a body-centred cubic structure. Above this temperature, iron changes to the γ-form, which has a face-centred cubic structure.

a) Draw the unit cells for the two crystalline forms of iron.

b) Describe, with explanation, the change in volume that would occur when the temperature of a piece of iron is increased from room temperature to 1 000 °C.

(HKALE 2004)

11 C_{60} is the most widely studied fullerene. It crystallizes in a closely packed face-centred cubic structure. The diagram below shows a representation of the unit cell of C_{60}.

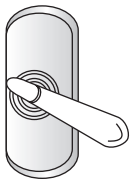
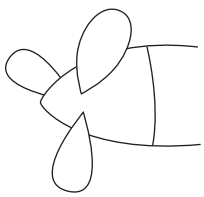


a) With reference to the diagram, calculate the number of carbon atoms in a unit cell of C_{60}.

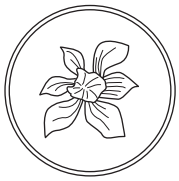
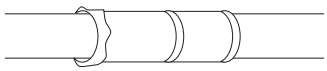
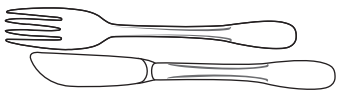
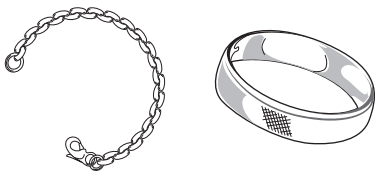
b) The face-centred cubic structure can be considered as a repetition of three different layers of closely packed C_{60} molecules. Mark, on the diagram, three different layers of C_{60} molecules in this cubic structure.

(HKALE 2005)

12 The table below gives the uses of some alloys. Complete the table by giving reason(s) for the use in each case.

Alloy	Use	Reason(s) for use
Brass	 making door knobs	• golden appearance • more corrosion resistant than pure copper
Bronze	 making ship's propellers	• more corrosion resistant than pure copper • resists wearing better than pure copper

Continued on next page

Alloy	Use	Reason(s) for use
Coinage metal	 making coins	• shiny appearance • more corrosion resistant than pure copper • harder than pure copper • malleable
Solder	 joining water pipes	• melting point of solder is lower than that of tin and lead • strong
Stainless steel	 making cutlery	• hard • strong • very corrosion resistant
Carat gold	 making ornaments	• stronger than pure gold • harder than pure gold

13 The table below shows the properties of some metals and alloys.

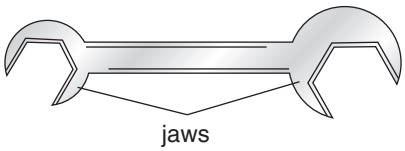
Material	Melting point (°C)	Hardness	Toughness
Copper	1 083	soft	easily bent
Lead	327	very soft	easily bent
Chrome-vanadium steel	1 500	hard	very tough
Wood's metal (an alloy of tin, lead, cadmium and bismuth)	70	soft	fairly easily bent

Decide which of these you would use for each of the following jobs. Write down the reasons for each choice.

a) To make the jaws of a spanner.

Chrome-vanadium steel

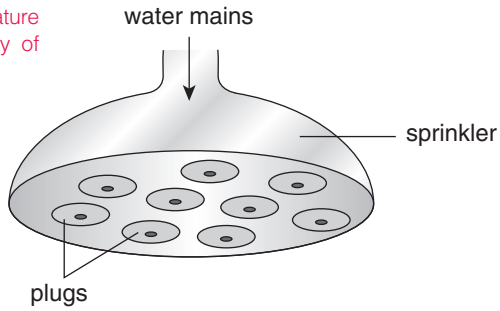
The material is very tough. In both the tightening and the undoing of nuts, the jaws are put under great strain. The jaws would deform and cannot grip the nut if the material were weak.



b) To make the plugs for an automatic fire sprinkler.

Wood's metal

The alloy plugs melt at a temperature above 70 °C, releasing a spray of water onto the fire.



Extension

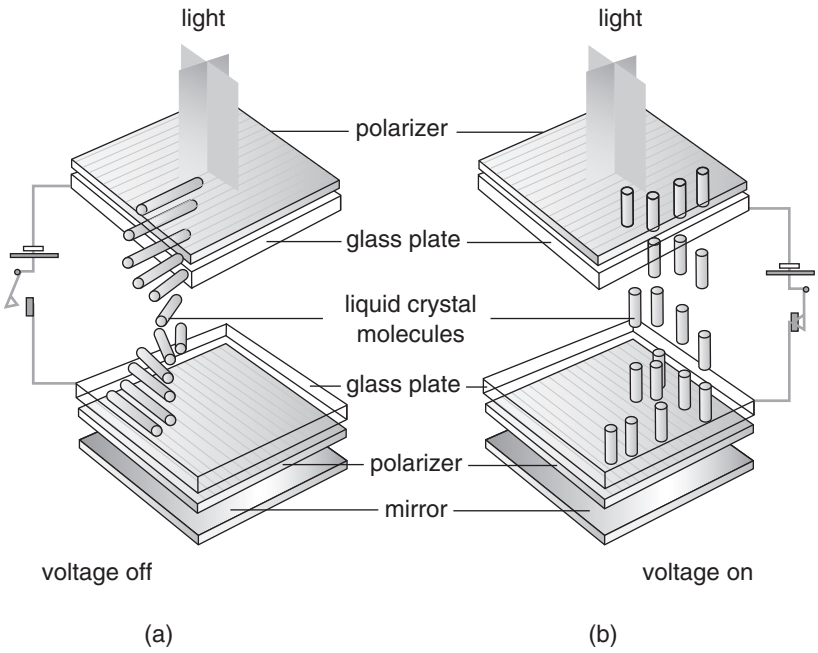
14 The properties of an alloy are different from those of the pure metals from which the alloy is made. Discuss the differences in properties with reference to their structures.

Refer to next page.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

15 The diagram shows the 'off' state and 'on' state of a typical liquid crystal display.

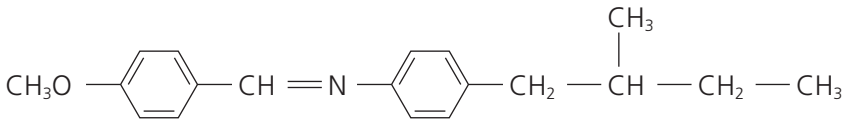
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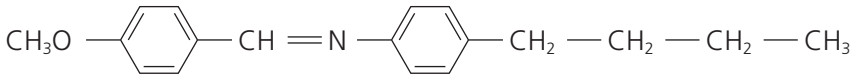
a) Explain why a white background appears during the 'off' state.

b) Explain why the area appears dark during the 'on' state.

16 Explain which of the following compounds, X or Y, can form cholesteric phase liquid crystals.



X



Y

17 Zinc oxide is a chemical used in some sunscreens.

a) What is the function of zinc oxide in sunscreens?
To absorb ultraviolet light which causes sunburn and damage to the skin.

b) What is the colour of zinc oxide?
White

c) What is the purpose of using nanoparticles of zinc oxide in sunscreens?
To make the sunscreens transparent.

14 The properties of an alloy are different from those of the pure metals from which the alloy is made of. This is due to difference in their structures.
In a pure metal, all the atoms are of the same size. The layers of atoms can slide past one another easily when a force is applied.
Hence a pure metal is both malleable and ductile.
In an alloy, atoms of a different size are added. This distorts the regular structure of the pure metal. The layers of atoms in the alloy are difficult to slide past one another when a force is applied. Hence the alloy is stronger and harder than the pure metals from which it is made. The alloy is also less malleable and ductile.
The melting point of a solid is very much dependent on its structure. A regularly packed solid has a higher melting point than one with a less regular structure. The structure of an alloy is less regular than those of the pure metals from which it is made. Hence an alloy usually has a lower melting point than the pure metals.
In an alloy, the presence of atoms of a different size hinders the movement of mobile electrons. Hence an alloy usually has a lower electrical conductivity than the pure metals.

15 a) The upper glass has a chemical treatment which makes the liquid crystal molecules close to it line up in a particular direction. The lower glass plate has been treated so that the liquid crystal molecules near it line up at right angles to the first direction.
The intervening molecules gradually rotate their relative orientation to accommodate the 90° change from one glass plate to the other. The light passing through the first polarizer will be guided by the molecules. Hence the plane of polarization of the light passing through the liquid crystal layer will be rotated through 90°.
As a result, the polarized light can pass through the second polarizer, strike the mirror and reflect back. Thus, a white background appears.
b) When a voltage is applied across the electrodes, it also applies an electric field across the liquid crystal. The liquid crystal molecules line up parallel to the field and no longer rotate the plane of polarization of the light.
Light cannot pass through the second polarizer to reflect off the mirror. Thus, the area appears dark.

16 Only chiral molecules can exhibit the cholesteric phase. X can form cholesteric phase liquid crystals as it is chiral.

